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Robocalls: A worldwide or US-only problem? Analyzing spam and fraud in international phone calls

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ABSTRACT

Unsolicited automated phone calls (robocalls) are a serious threat: in the US alone, these calls resulted in reported losses of $1.1 billion during 2025. Phishing and spoofing consistently rank among the most reported crimes within the FBI's Internet Crime Complaint Center, with phone call scams having the highest reported median loss. Combating robocalls is difficult due to many legal and practical constraints: robocalls often encompass multiple legal jurisdictions of different countries/states, the large volume of robocalls, their multilingual nature, the lack of publicly available data, privacy concerns with obtaining data, etc. We present a study of international robocalls, aggregating robocall reports from countries across all inhabited continents and contribute by providing new findings on international robocalls from 65 different countries. We also present the first publicly available multimodal and international robocall dataset: 8.7 million call detail records, 839 robocall transcripts from 28 identified robocall campaign clusters, and 677 robocall recordings. We describe our methodology for collecting robocall data over a 9-month period and provide a detailed analysis comparing robocalls in the US with those in other countries. Our analysis covers several aspects, including uncovering calling patterns, analyzing temporal co-targeting groups, discovering common robocall campaigns, extracting callback numbers, analyzing linguistic differences among robocalls in the same language but different regions, and other insights. Our results indicate that robocalls are an international problem with substantially higher call rates for US numbers than for pooled non-US numbers in our honeypot.

1. Introduction

Fifteen years ago, most readers would probably have been unfamiliar with the term *robocall*, and the broader public would likely never have personally encountered a robocall. This is no longer the case as many people worldwide, especially in the US, are continuously exposed to robocalls, and the problem shows no signs of slowing down. Fig. 1 shows the average number of monthly reported robocalls from 2019 through 2025. These numbers were obtained from the U.S. Public Interest Research Group Education Fund (2025) and, as can be observed, robocalls have been on the rise over the past three years and reached a six-year high in 2025. In December 2025 alone, a total of 4.1 billion robocalls were made to US consumers, with the volume of scam and telemarketing calls increasing by 15.6% in 2025 compared to 2024 (U.S. Public Interest Research Group Education Fund, 2026).

1.1. What exactly are robocalls?

Definitions vary. According to the US FCC (Federal Communications Commission, 2025): "*robocalls are calls made with an autodialer or that contain a prerecorded or artificial voice message*". Prasad et al. (2020) define robocalls as a *catch-all term for automated or semi-automated unsolicited calls, often for fraud or telemarketing purposes*. Some authors refer to robocalls as "*computer-generated voices*" (Elizalde and Emmanouilidou, 2021).

Not all robocalls are necessarily malicious or illegal. Some are even highly desirable, such as public safety announcements or school closure announcements (Federal Communications Commission, 2025; Prasad et al., 2020). In the US, telemarketers are allowed to make calls using prerecorded or artificial voices, provided that they have obtained prior express written consent from the callee and that these calls are made between 08:00 and 21:00 (Federal Communications Commission,

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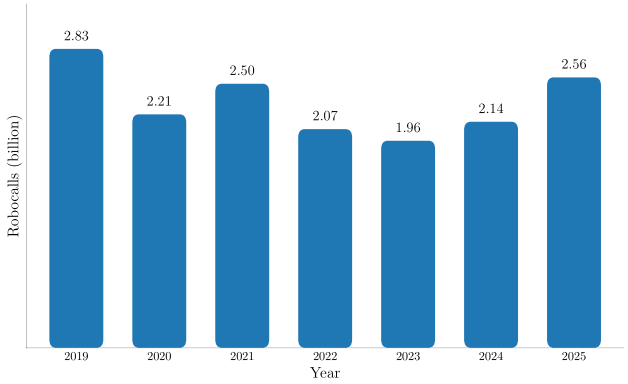


Fig. 1. The average number of monthly robocalls in the US over 2025 was 2.56 billion, which is the highest since 2019.

2025). The FCC's website contains more information regarding the legality of these calls in the US.

Robocalls are not a minor inconvenience; in fact, illegal robocalls are the FCC's top consumer complaint and top consumer protection priority (Federal Communications Commission, 2025). According to the US Federal Trade Commission (FTC), in 2025, US citizens have reported a total of $1.1 billion lost to fraud over phone calls (Federal Trade Commission Public Tableau, 2025a). Therefore, stopping robocalls is of interest to the general public.

Given that the two definitions of robocalls provided by the Federal Communications Commission (2025) and by Prasad et al. (2020) are highly aligned, in this paper, we will consider any unsolicited calls that contain a prerecorded or artificial voice message as robocalls. Such calls do not necessarily have to be malicious. We will provide more details and a stricter terminology in the following section.

1.2. Why now?

There are several reasons why robocalls are an ever-growing issue, but they all more or less point to the fact that it is easier than ever to create robocalls and that it is difficult to trace them, thereby making legal prosecution difficult:

- The telephone network is still to date (2025) the largest network of human users, with more active users than the internet and more mobile-cellular subscriptions than the Earth's population (International Telecommunications Union Telecommunication Development Sector, 2024, 2026). This means that malicious users have a large pool of targets to choose from.
- It is difficult to regulate and trace malicious calls, as calls can be made from any place on Earth while targeting users worldwide. This makes legal action more difficult as it requires international cooperation, and, to complicate matters further, robocalls are regulated differently depending on the laws and regulations of each country. The low chance of being caught makes it appealing for malicious users to attempt fraudulent behavior.
- The telephone network operates in such a manner that the actual originating number does not have to match the caller ID which is displayed on the callee's device. This is known as *caller ID spoofing* and will be discussed in more detail in the paper.
- The rapid advancements in technology, most notably AI, have made it very easy to generate artificial voices that sound very human-like (Cui et al., 2025). Text-to-speech technology and translation tools enable malicious users to synthesize audio in languages they do not even understand.
- Similar to the previous point, advancements in technology, such as coding/programming assistants (Jiang et al., 2026), now enable users with no technical or programming expertise to interact with technology, specifically software and APIs that require(d) programming knowledge. This makes it easy for them to leverage cloud communication providers for malicious acts.
- Creating calls is cheap, especially since unanswered calls are not charged at all.
- Finally, such fraudulent behavior has a track record of being successful, as millions have already been stolen from people, making the misbehavior appealing to malicious users (Federal Communications Commission, 2025; Federal Trade Commission Public Tableau, 2025a).

1.3. Research questions

This paper addresses the following research questions:

- **RQ1**: How does robocall volume differ between US-targeted and non-US-targeted traffic?
- **RQ2**: How do temporal calling patterns and caller concentration differ by region, and do co-targeting groups exhibit similar call content?
- **RQ3**: How do the robocall scripts differ by region and language and what are the most common robocall types per region and language? Do robocalls differ by audio content, e.g., are spoofed robocalls equally frequent in the US and other regions?

1.4. Paper contributions

Among others, the main contributions of our paper can be summarized follows:

- We provide the first international and multilingual analysis of robocalls. To the best of our knowledge, existing literature on the topic of robocall analysis mostly addresses US and North American data, with no available analysis of robocalls within Africa, Asia, Australia, Europe, or South America. Naturally, this means that existing literature focuses on robocalls that are mostly in English language.
- We set up a honeypot consisting of more than 100 thousand phone numbers with the intent of recording robocall activity. These phone numbers are international, and, to the best of our knowledge, represent the first international study of robocall activity.
- We publish the first open dataset containing more than 8.7 million call detail records, recorded by our honeypot. We are not aware of any prior open datasets of call detail records.
- We contribute by making available 677 robocall audio recordings and 839 robocall transcripts. These recordings and transcripts are in: English, Spanish, Chinese, and Polish and have been manually verified by at least one human annotator to be robocalls. We are only aware of one other dataset of audio robocalls (Prasad and Reaves, 2023), which contains robocall audio recordings from the US published by the FTC.

2. Background

Nassar et al. (2007) started their 2007 work with the sentence: "*Even if VoIP attacks are not in the headline news in security reviews yet, they soon will be of major harmfulness for the Internet telephony services*". VoIP is short for Voice over IP, the technology that enables propagation of calls over the internet, and unfortunately, as we can witness today, the prediction of the authors lived to be true.

Prasad et al. (2020) provide a great overview of how the phone network operates, which we briefly summarize as well. When a caller

dials a number, they are said to *originate* the call, while their carrier is responsible for establishing the connection, which is poorly named as *terminating* the connection. If the dialed number is not directly accessible to the carrier, the carrier routes the call by sending call signaling and media through one or more intermediate carriers. These may or may not operate in the same country. The most common signaling protocols are: Signaling System No. 7 (SS7) (International Telecommunication Union Telecommunication Standardization Sector (ITU-T), 1988) for the Public Switched Telephone Network (PSTN) and Session Initiation Protocol (SIP) (Schooler et al., 2002) for Voice over Internet Protocol (VoIP). Carriers connect through network gateways that can translate between signaling protocols.

Furthermore, Prasad et al. (2020) explain the underlying mechanisms of how *spoofing* is performed. Spoofing is a cyber tactic in which the perpetrator is identifying as another person or program, which in the context of robocalls means effectively falsely displaying the caller's number (caller ID) to the callee and is known as *caller ID spoofing*.

An intuitive first reaction to the problem of robocalls might be: "why not simply block the malicious user's number from making calls?". The problem is that calls can be spoofed, making it very difficult to trace the actual number of the perpetrator. SS7 is a protocol designed in 1970, standardized since 1988, and it is still in use. In hindsight, it seems logical to implement a type of authorization, but neither SS7 nor SIP have built-in authentication of the calling number. This is because historically it was assumed that only trusted carriers would terminate calls. Aside from the obvious reason of hiding their true identity, robocallers spoof numbers to make the call appear to originate from a legitimate source.

STIR (Secure Telephone Identity Revisited) (Peterson et al., 2014; Peterson and Turner, 2018; Rescorla and Peterson, 2021) is a series of Request for Comments (RFC) standard documents by the Internet Engineering Task Force (IETF). It enriches the SIP protocol by adding digital certificates to verify the caller's ID. Similarly, SHAKEN (Signature-based Handling of Asserted information using toKENs) (Wendt and Barnes, 2019) is a set of guidelines for the PSTN on how to deal with calls that have incorrect or missing STIR.

In 2020, Prasad et al. (2020) were optimistically looking forward to STIR/SHAKEN as the first and only widespread cryptographic authentication mechanism in telephony. The FCC has enforced STIR/SHAKEN as of June 30, 2021 through the TRACED Act (Telephone Robocall Abuse Criminal Enforcement and Deterrence) which was passed by the US Congress in late 2019. Similarly, The Canadian Radio-television and Telecommunications Commission instructed telecommunication service providers (TSPs) to implement STIR/SHAKEN effective November 30, 2021 (Canadian Radio-television and Telecommunications Commission, 2021).

These mechanisms have, frankly, not been effective. Prasad et al. (2020) published new findings (Prasad et al., 2023, 2025) in which they provide explanations as to why STIR/SHAKEN have failed to reduce the volume of robocalls. In essence, STIR/SHAKEN cannot completely authenticate calls originating outside a country or calls that transit a legacy phone circuit (Prasad et al., 2023), which robocallers have adapted to Prasad et al. (2025).

In general, spoofing can be summarized as faking one's identity. The process we just described in which callers fake their identity is known as *number spoofing* or *caller ID spoofing* and should not be confused with *voice spoofing* (Rosello et al., 2023; Tran et al., 2024; Mahapatra et al., 2025; Das et al., 2025). The latter refers to malicious users applying AI technology to clone (spoof) a trusted voice (e.g. the voice of a family member) and trick the callee into believing they are communicating with somebody they trust, thereby potentially extorting valuable information, money, or obtaining system access. Finally, *voice phishing*, also known as *vishing* (Lee and Park, 2023; Ampel et al., 2026), is a social engineering attack that is carried out over the phone, which can, but does not necessarily include voice or caller ID spoofing. During a vishing attack, the perpetrator tries to trick the victim over the phone to obtain information, money, or to bypass authorized access. Vishing attacks, caller ID spoofing, and voice spoofing can all be considered robocalls, as per the definition of the FCC, provided that they were made with an autodialer or contain a prerecorded or artificial voice message.

3. Related work

Robocalls are not a recently emerging issue. Although the term *robocall* might be relatively novel, unsolicited and malicious phone calls have been well-known for some time. Researchers from North Carolina State University (NCSU) provided a substantial contribution and effectively laid the foundational work on the topic of robocalls. Over the years, they published several papers on the topic of robocalls, and more generally on voice and SMS security (Prasad et al., 2020, 2023, 2025; Reaves et al., 2016b,a, 2017; Nahapetyan et al., 2024; Adei et al., 2024). A limitation of their work is that it considers US data only.

The systematization of knowledge paper by Tu et al. (2016) provides an overview of robocalls up to that point in time. At the time of writing, robocalls were an increasing issue as well, with an annual 22 million complaints made to the FTC about illegal calls. The paper focuses predominantly on the US.

A recent 2025 paper by Pietri et al. (2025) provides a more up-to-date overview of telecom spam and scam, from a perspective of the advances in AI. The paper provides a perspective on several countries across all continents. As an example, the authors show that South Africa, Nigeria, Kenya, and Ghana, have made advances such as: passing bills that mandate TSPs (Telecommunications Service Providers) to verify user identities, requiring that telecom operators implement filters/systems to block spam calls, or enforcing that TSPs maintain a registry for consumers to opt out of unsolicited communications.

Agarwal et al. (2026) propose the *AI Regulatory Readiness Index* (ARRI), a reproducible instrument for doctrinally assessing the legal preparedness of national frameworks to govern AI in critical digital infrastructure. They apply the ARRI across ten jurisdictions spanning five continents and find that global AI regulatory readiness in telecommunications remains concentrated in the lower range. Traditional cybersecurity frameworks, such as those from the ITU (International Telecommunications Union Telecommunication Standardization Sector, 2012; International Telecommunications Union Telecommunication Development Sector, 2022; International Telecommunications Union Telecommunication Standardization Sector, 2024), primarily address conventional threats, while overlooking risks inherent to AI systems.

3.1. Honeypot methods

Honeypots, in general, are systems designed to record undesirable user behavior or, in certain cases, even to attract such behavior, hence the name *honeypot* (Provos et al., 2004). They have no production value and are based on the assumption that any interaction with them is unauthorized, which means that the traffic that they receive is usually a probe, scan, or attack (Spitzner, 2003). A honeypot may have different definitions depending on the use case, but in the context of robocalls, it can be considered as a collection of phone numbers that deliberately answer incoming phone calls from unknown callers, as these are more likely to originate from robocallers. In the late 2000s and early 2010s, designing VoIP honeypots was a popular research topic (Gupta et al., 2015; Nassar et al., 2007; Jiang et al., 2013; Keromytis, 2012; do Carmo et al., 2011; Mulliner et al., 2011; Nawrocki et al., 2016).

In their 2004 paper, Pouget et al. (2004) used a honeypot to, among other things, reduce spam activity and deceive hackers. Honeypots were utilized by Google (Provos et al., 2004), researchers from Deutsche Telekom and TU Berlin Mulliner et al. (2011), and researchers from the NCSU (Prasad et al., 2020, 2023, 2025).

Gupta et al. (2015) monitored 39,696 phone numbers. They received a total of 1.3 million calls from 250 thousand unique sources

over a period of seven weeks and detected several debt collectors and telemarketer calling patterns and one instance of a telephony denial-of-service attack. The authors observed that the weekday call volume (≈33,000) was higher than the weekend call volume (≈8,000), while during weekdays calls were more likely to occur during business hours. The study was conducted in the US and found that most calls were targeting the US state of California. The honeypot interacts with callers by playing an interactive voice response (IVR).

We already referenced several aspects of the 2020 paper from researchers of the NCSU (Prasad et al., 2020), given that the authors provide explanations of relevant concepts such as robocalls, spoofing, STIR/SHAKEN, and similar. This paper was at the time of writing the first paper in which a large-scale honeypot was employed. The authors recorded unsolicited calls to a honeypot of up to 66,606 lines over 11 months. The paper features a detailed analysis of the Call Detail Records (CDRs) for all calls and call content. CDRs contain a brief summary of the call, such as callee and caller number (which could be spoofed), completion status, start and end timestamps, etc. During the 11-month recording period, the authors collected 1,481,201 unsolicited phone calls. They provide further details on how they processed a large number of audio files by leveraging audio processing techniques, mainly silence detection (voice activity detection) and audio fingerprinting.

In their 2023 paper (Prasad et al., 2023), researchers from NCSU collected 232,723 robocalls over a 23-month period. They present a framework called *SnorCall* which enables scalable and efficient extraction of content from robocalls. In a 2025 continuation (Prasad et al., 2025), the authors tackle a significantly larger set of about 3 million voice calls. The main finding of this paper is that robocalls were in decline (which was true at the time), but that robocallers had managed to adapt to STIR/SHAKEN.

Carrillo-Mondéjar et al. (2022) employ a VoIP honeypot that is running on their custom infrastructure using Asterisk to build their private branch exchange (Van Meggelen et al., 2019; Sangoma Technologies, 2026). This work focuses on how scammers obtain infrastructure or resources to make calls at no cost and what type of actions they perform by monitoring a real VoIP system.

Prasad et al. (2025) analyze data from multiple independently operated vantage points, ranging from industry and academic honeypots to public enforcements and consumer complaints. As an example, they utilized RRAPTOR (ZipDX LLC, 2025), an interactive commercial honeypot that automatically answers received calls and plays a set of pre-recorded prompts. Another example is robocall observatory, a non-interactive honeypot that answers calls with a fixed greeting message: "Hello", after which it simply records the call. The authors found that large-scale and long-running honeypots are reliable vantage points to measure broader robocalling phenomena. Among other, the authors provided new findings such as: (1) automated audio-based campaign identification is more specific than human-assigned labels, (2) interactive honeypots capture nearly 10 times more callback numbers from robocalls, and (3) voicemail injection techniques were used by 362 robocall campaigns to deliver messages to the target's voicemail.

MobiPot (Balduzzi et al., 2016) is the first mobile phone honeypot that over a period of 7 months collected more than two thousand voice calls and SMS messages, half of which were unsolicited. The honeypot was deployed in China and features a semi-interactive operational mode; authors emulate a receiver facing communication difficulties to keep the attacker more engaged by playing pre-recorded messages such as: "Hello?", "Do you hear me?", and "Better now?".

3.2. Voice spoofing attacks

Voice spoofing attacks are malicious attacks whereby an adversary attempts to masquerade as another individual through the presentation of manipulated speech data (Liu et al., 2023), with four principal means of attack generation: *impersonation*, *voice conversion (VC)*, *text-to-speech (TTS) synthesis*, and *replay*. The first method requires special expertise, whereas VC, TTS, and replay can be performed with readily available software toolkits and consumer devices.

Rapid advances in generative modeling have enabled easy generation of synthetic speech, thereby narrowing the distinction between authentic (real) and artificial (synthetic) speech resulting in speech-based services being vulnerable to spoofing attacks (Das et al., 2025). Yu et al. (2024) highlight the risks of AI-powered social engineering attacks, citing the 2023 FBI report which attributes close to $1 billion of loss to AI-powered scams. Pantserev (2020) discusses the malicious use of AI-based deepfake technology and how it poses a new threat to the psychological security and the political stability.

ASVSpoof (Liu et al., 2023) is a spoofing and deepfake detection initiative and challenge series. In the ASVSpoof 2021 challenge, authors analyze performance of spoofing detection models with respect to the underlying audio codec used, which is G.711 for telephony (International Telecommunication Union Telecommunication Standardization Sector, 1988) (the international standard for PSTN). An observation from the paper is that narrowband audio, such as that produced by G.711 results in worse performance of anti-spoofing models, indicating the importance of higher frequency content for accurate spoofing detection. Spoofing-detection models are also sensitive to silence (Zhang et al., 2023): removing the silence from test datasets using voice activity detection (VAD) severely degrades performance, attributed among others to the fact that TTS generated audio contains a lower proportion of silence compared to bonafide speech.

AASIST3 (Borodin et al., 2024) is a state-of-the-art anti-spoofing model that extends upon the AASIST architecture (Jung et al., 2022) by enhancing it with Kolmogorov–Arnold networks (KAN) (Kolmogorov, 1957). The latest model, Spectra-AASIST3 (Lab260, 2026), was released in 2026 and provides an anti-spoofing model that pairs XLS-R-300m (Babu et al., 2022), a self-supervised speech model based on wav2vec2.0 (Baevski et al., 2020), with the KAN-enhanced AASIST model and achieves the best results across various benchmarks.

3.3. Detecting fraud in telephony

Sahin et al. (2017) presented a bot named *Lenny* that answers inbound calls and can interact with callers using a set of prerecorded voice messages. It was capable of conversing for tens of minutes while only being clearly recognized as a bot in 5% of the calls. Similarly, Pandit et al. (2021) detected mass robocalls using a virtual assistant as well. The assistant intercepts incoming calls and asks callers to identify themselves before forwarding the call to the callee, achieving an accuracy of about 93%. *RoboHalt* (Pandit et al., 2023) is a continuation of Pandit et al. (2021); the authors design a virtual and interactive assistant named *RoboHalt* that answers inbound phone calls on behalf of the callee and holds a natural conversation with the caller. It relies on multiple NLP (natural language processing) machine learning models to detect whether the caller is a human or a robocaller and works regardless of whether the caller is employing caller ID or voice spoofing. The system achieves a combined accuracy of 88.7% on various types of robocalls. Virtual assistants should be considered with caution, as research has found that they introduce other security risks (Bolton et al., 2021), with reports of sophisticated attacks being able to extract personally identifiable information from virtual assistants. Malicious attacks are performed in varying forms (Mitev et al., 2019): artificially synthesized speech (Diao et al., 2014), obfuscated speech (Carlini et al., 2016; Vaidya et al., 2015), embedded voice commands (Schönherr et al., 2019; Yuan et al., 2018), or ultrasound-modulated voice commands (Zhang et al., 2017).

Telesonar (Yan et al., 2022) is an AI-powered robocall detector, that relies on echo channel detection and breath sound timing, used to determine whether speech is AI-generated or human. Elizalde and Emmanouilidou (2021) proposed a method for discerning between computer-generated and human-generated voices, as well as between

spam and non-spam audio. They compare methods that directly operate on spectro-temporal features against methods that operate on features extracted using the OpenSmile library (Eyben et al., 2010). In both of these approaches, robocalls are only considered to be AI-synthesized speech, while prerecorded messages do not count as robocalls.

Natarajan et al. (2021) transcribe calls and convert them into a high-dimensional latent space, after which they classify them as either spam or ham. Lee and Park (2023) proposed a model for real-time vishing detection in Korean, which is also based on the transcript of the call.

Kapourniotis et al. (2011) present a fraud detection framework for VoIP calls. They use a Bayesian network to determine whether a call is fraudulent or not by only analyzing CDR data. Similarly, Jiang et al. (2012) present a method for detecting voice-related fraud activities using only call detail records. Their approach uses *voice call graphs*, which represent voice calls from domestic callers to foreign recipients, and propose a Markov clustering-based method for isolating dominant fraud activities from international calls. They collected data over a two year period from one of the largest cellular networks in the US.

Tas and Baktir (2024) propose a blockchain-based caller-ID authentication method, that would prevent caller ID spoofing in real-time. A similar approach that utilizes the blockchain for identity authentication was proposed by Chen et al. (2021). Tseng et al. (2015) proposed FrauDetector, a graph-mining-based framework for fraudulent phone detection. Although the paper does show promising results, it does not take into account spoofing attacks.

Research on attacks facilitated through SMS is closely related to robocalls (Reaves et al., 2016b; Nahapetyan et al., 2024; Jiang et al., 2013; Mishra and Soni, 2020; Reaves et al., 2018; Abdulhamid et al., 2017). What makes robocalls more challenging is Tu et al. (2016):

1. *Immediacy.* Call requests are immediate, and therefore anti-spam systems must complete analysis within a short window of time.
2. *Audio streams.* The spoken content of a call is not known in advance, unlike in text-based communication, therefore, any content-based analysis must be performed in real-time in a streaming fashion, while detection methods must be causal.

4. Method

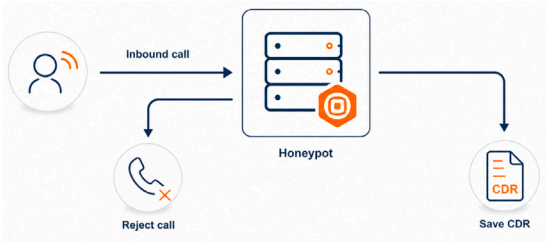
In this section, we describe the method that we used to record robocall activity. All calls were recorded using Infobip,[1] an international communication platform. Infobip provided us with the phone numbers and hardware to record the calls, under certain ethical agreements and legal obligations that we further discuss in this section.

Fig. 2 provides a simplified diagram of two different honeypot configurations. We used the same honeypot consisting of the same phone numbers, but in two different configurations (setups/modes).

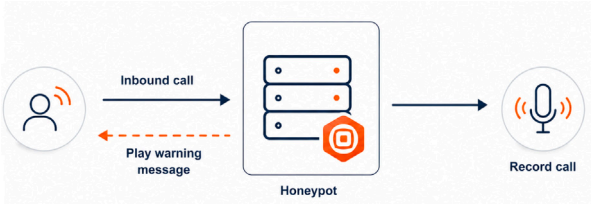
The first setup is termed *passive*, which corresponds to Fig. 2(a). In this scenario, any incoming calls are simply rejected by the honeypot, and the honeypot does not interact with the calls. However, the honeypot registers the basic details of the call in a database. These are the previously mentioned CDRs, which contain basic call information (caller ID, callee ID, timestamp, and duration – in this case, always 0).

The second honeypot setup is the *interfering setup*, shown in Fig. 2(b). In this scenario, the honeypot answers the incoming call and plays a warning message (an announcement/notification) to the caller, based on the caller's country prefix, to inform them about the recording taking place.

The honeypot was active from 6 June 2025, until 14 February 2026, including one longer and one shorter period of inactivity. The first period of inactivity occurred during summer, i.e., 1 July 2025 until 15 September 2025, because of a lack of personnel who could monitor the honeypot. A short 10-day period of inactivity occurred during late



(a) Honeypot method 1: passive (non-interfering mode). Incoming calls are simply rejected, but the caller ID, destination phone number, and timestamp are registered and added to the database.



(b) Honeypot method 2: interfering mode. Based on the caller's country, a warning message is played to them in their country's official language to inform the calling party that the call is being recorded. Callers are given enough time to hang up to avoid being recorded.

Fig. 2. Two honeypot configurations used for collecting data.

November, a well-known holiday period (Thanksgiving, Black Friday, Cyber Monday), as we could not guarantee the stability of the honeypot during that period given the expected increase in call volume. The number of calls received by the honeypot per day is shown on Fig. 3. Calls registered while the honeypot was running in passive mode are marked in blue, while calls registered while the honeypot was running in interfering mode are in orange.

4.1. Call detail records

When operating the honeypot in its passive mode, we only register the caller's ID and country (and state, for US calls). Furthermore, we assign a unique ID to each caller ID and perform all downstream tasks on those unique IDs. This means that we can still analyze and differentiate between callers, but do not know their true identity (instead of operating on actual phone numbers, we operate on unique IDs). This is a safety mechanism to ensure that even if our honeypot receives legitimate calls from live humans, their caller ID will not be revealed. Furthermore, after consulting with our legal and ethics staff, we agreed to delete all call records for which the unique ID occurred only once, as these are more likely to have originated from live humans who potentially misdialed a number from our honeypot.

To the best of our knowledge, the anonymization scheme that we used does not permit recovery of a caller's or callee's true number from the unique IDs. Rather than encrypting or hashing phone numbers, we assign each phone number a randomly generated UUID. We deliberately avoided hashing: because phone numbers are a small, effectively enumerable space, a malicious party could hash every number of interest and match it against our data, thus hashing alone would not provide meaningful anonymity. Randomly generated UUIDs with no deterministic relationship to the underlying number avoids this weakness. For simplicity, we will hereafter refer to the unique IDs as the caller ID.

[1] https://www.infobip.com/.

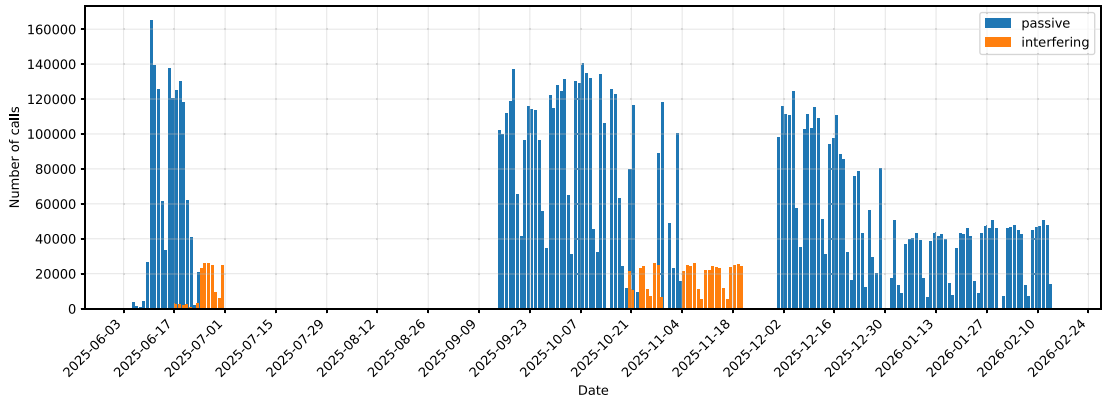


Fig. 3. Daily volume of phone calls registered by the honeypot. Days colored in blue represent calls registered while the honeypot was operating in *passive* mode, i.e. the honeypot does not interact with the calls, it simply rejects them. Days colored in orange indicate the *interfering* mode, i.e. the honeypot answers calls and plays warning messages to inform callers about the recording process. We noticed a pattern that interacting with robocalls resulted in substantially fewer robocalls. This may imply that robocallers are analyzing the response type, which we consider in detail later. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Call audio recordings

The interfering mode of our honeypot is more delicate from an ethical and legal standpoint, as it requires recording calls. For a majority of countries, it is sufficient for one party of a phone call to consent to recording for the recording process to be legal. However, to minimize any potential risks, we still explicitly inform the calling party about the recording process and provide them sufficient time to hang up, before the recording takes place. We play a warning message in the language of the caller's country to warn them that the call will be recorded. If the caller does not hang up within 5 s, the call is recorded. The template message played to the calling parties is the following:

> "You have dialed a number that is not in service. This call is being recorded to prevent spam and robocall scams over the telephone network, and the recording may be shared with partners for this purpose. If you do not want this call to be recorded, hang up within the next 5 s."

Despite the warning message, we still introduce four additional safety mechanisms:

1. No metadata about recordings will be published. The recordings are anonymized and assigned a unique, randomly generated ID, using a similar process as the CDR anonymization scheme. Recordings cannot be traced back to any caller ID or caller country.
2. Any published recording must be verified by a human. The number of robocall recordings that we make publicly available (677) is significantly smaller than the number of CDRs (more than 8.7 million). If a recording is included in our dataset, it is confirmed by at least one human evaluator to be a robocall prior to publishing.
3. All other (unverified) recordings were filtered in a process we describe later to minimize the chance that they are accidental calls made by humans. They were used for automated computer analysis and deleted afterwards, ensuring that they can no longer be accessed and were only used for the purposes of analysis in this paper.
4. Finally, any recordings shorter than 5 s were automatically deleted to further ensure that accidental misdials are excluded from our research.

Residual risk mitigation. The safeguards described above cannot fully rule out that a small number of legitimate, human originated calls were captured and mislabeled as robocalls by an annotator. We mitigate this residual risk by committing to a takedown process: the dataset's terms of use obligate downstream users to report any recording they believe originates from a legitimate human caller, so that it can be removed from the public release upon verification.

4.3. Legal and ethical considerations

Similar to previous experiments involving honeypots, we had to consider and comply with several ethical and legal requirements (Prasad et al., 2020, 2023, 2025; Gupta et al., 2015; Carrillo-Mondéjar et al., 2022; Balduzzi et al., 2016; Sahin et al., 2017). From an ethical standpoint, our ultimate objective is to advance the knowledge of robocalls, and hopefully, contribute to a more reliable and secure telephone network. All content that we release will be publicly available for non-commercial use. We consulted with our legal department and several ethics boards in Austria and Croatia to ensure that our work is ethical and within valid legal frameworks. We have not advertised our robocall honeypot, nor did we make outbound calls. Ideally, this would mean that we should not receive any calls to our honeypot except from robocallers running mass campaigns. However, it can be the case that an individual either calls one of the available numbers in the honeypot or that a number has changed ownership in the meantime, thus we cannot assume that all calls the honeypot receives are robocalls, even though a large majority will be.

4.3.1. Consultation with ethics and legal staff

The main concern that we had while discussing with our ethics and legal staff was ensuring that our study remains within ethical and legal boundaries.

Together with our team, we consulted related work to analyze the considerations and dilemmas that earlier work had dealt with. For example, Prasad et al. (2020) also use a honeypot method to collect calls and acknowledge the risk of a legitimate call from a human being collected by the honeypot. They state that the principal data collection methodology is to wait for the arrival of calls on inbound lines owned by the authors. In such a setting, the authors are neither targeting nor recruiting participants to take part in their study. The authors do not reach out to any participants and refrain from advertising the phone numbers in any portals, social media, or through any other mechanisms. As such, the study is similar to research studies that perform public observations, with the exception of their observation taking place in a virtual environment. Prasad et al. (2023) provide similar ethical considerations, but also suggest that they do not perform

steps to traceback or positively identify callers through any means. If transcription services were used, the authors ensured that the audio would not be shared or used for training models, for example by paying premium services.

Our methodology is aligned with these considerations, with the addition that we also inform callers of the recording taking place and provide them sufficient time to end the call. This consultation is a concrete example of a design trade-off: as shown in the distribution of call volume across honeypot modes (Fig. 3), switching to the interfering mode and playing the warning message caused call volume to drop substantially, in some cases by an order of magnitude. We accepted this loss of data volume and completeness in exchange for uniform, conservative call handling across all jurisdictions.

Regarding legal considerations, our legal team advised us to inform the caller about the recording taking place, regardless of whether in some countries single party consent was sufficient. From a scientific perspective, this would also make the data more credible as the recordings would be obtained under the same conditions for all countries (i.e., the caller is informed about the recording taking place regardless of their country). Also, although our understanding is that some countries enable more flexible recording processing for fraud/spam prevention purposes, given that we are not directly preventing spam/fraud, but hopefully making a contribution towards preventing spam/fraud, our legal team still advised that we inform callers of the recording taking place and provide sufficient time for the caller to hang up.

Our dataset is hosted on Zenodo, operated by CERN. It is publicly accessible and licensed under Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). We ask that it be used only for combating fraud and spam robocalls or for research aimed at better understanding robocall behavior.

4.4. Transcripts

Similar to the recordings, we release a collection of 839 call transcripts. These were verified by at least one human annotator to be robocalls and have been processed to remove any personal identifiers. Call transcripts are released without accompanying metadata, so they cannot be traced back to the caller. Transcripts are released as a part of our automated robocall clustering and campaign analysis, which we cover in the next sections.

4.5. Limitations

Our honeypot has three main limitations:

- **Warning messages.** Playing the warning message may influence robocallers, e.g., robocallers might be reluctant to call the same or similar numbers once aware of the recording. We consider this effect in more detail later on.
- **Periods of inactivity.** Periods of inactivity interrupt the continuity of the honeypot. It would be better if our data were continuously collected throughout the entire recording process.
- **Variability of available phone numbers.** A technical limitation of our study was that the quantity of available phone numbers varied slightly per day. This variation is relatively small, on average 2.66% for US telephone numbers and 0.05% for all other phone numbers. More details are provided in Table 1, which lists the information for the US and top 15 countries by number quantity. The full list of numbers per country is available in the dataset.
- **Honeypot representativeness.** The phone numbers used in our honeypot were numbers previously assigned to messaging/voice services that had no active subscriber traffic for at least 6 months. Number counts per country (Table 1) therefore reflect the honeypot provider's number availability in each market, not a stratified or population-proportional sample, implying a potential bias. We consider this a technical limitation of the study that is shared with related work and is difficult to bypass, especially when taking into consideration that this is the first study of robocalls at an international scale. If we wanted to have a completely unbiased sample of global telephone numbers inside our honeypot, we would need to orchestrate the honeypot to operate from multiple telecom operators in each country, proportional to their geographic location and market share. Thus, we explicitly note that the numbers in our honeypot are not a random sample of the global telephone number space. Because the numbers are inactive/unassigned to a live subscriber, they cannot receive calls from parties who intend to reach a specific known person. Thus, any traffic they receive is a result of either mass/automated dialing that does not depend on the number being previously known to the caller, or a misdial of a number. This design ensures that the collected traffic is an indirect measure of the indiscriminate way robocall campaigns are orchestrated. However, this design will also underrepresent campaigns that rely on curated or purchased numbers lists of known/actives subscribers, i.e., targeted, human-orchestrated, individual attacks. We consider this an acceptable trade-off since our main goal is analyzing whether there is a discrepancy between the volume of mass orchestrated/automated robocall campaigns in the US compared to the rest of the world.

Overall, we approached the sensitive topic of collecting robocall data with extreme caution and have taken several steps to ensure no violation of caller privacy and to minimize the chance of accidental/live calls made by humans being caught by our honeypot.

5. Call metadata analysis

This section presents the analysis of robocall metadata collected from our honeypot infrastructure. A key consideration to keep in mind is caller ID spoofing. Robocallers routinely falsify the displayed phone number, so a call may appear to originate from the callee's country, to increase the level of trust. Because caller metadata is unreliable, we anchor our temporal and geographic analyses on callee location, the known locations of our honeypot numbers, framing results in terms of when and where robocallers target victims rather than where they operate from.

5.1. Call patterns

In this subsection we show some of the most prominent call patterns, such as the volume of received calls, the average daily call distribution, and the call distribution among the available honeypot numbers.

5.1.1. Dataset overview

The honeypot network received 9,656,257 raw CDRs between June 2025 and February 2026. We applied two preprocessing filters to isolate robocall activity: (1) we only selected calls which the honeypot had rejected (passive operating mode), as answered calls (interfering operating mode) were processed separately for audio analysis; and (2) we retained calls from phone numbers that appeared at least twice in the dataset, filtering out likely accidental human dials.

The resulting dataset contains 8,747,351 calls from 573,823 displayed caller IDs to 249,819 callee identifiers. It spans 254 calendar dates of which 163 contain retained CDRs and 91 are unobserved. Across the full observed period, the retained-call count corresponds to 34,438 calls per calendar day. Note that this value takes into account 91 unobserved dates.

In the analysis dataset, 8,314,813 calls target US numbers (95.1%) and 432,538 target non-US numbers (4.9%). For exposure adjustment, we matched calls and daily inventory from 7 October 2025 through 14 February 2026 using UTC dates. Of 131 calendar dates in total, 116

Table 1
Quantity of phone numbers employed in the honeypot. Although the total amount of available phone numbers varied per day, this variance manifested steadily and on average resulted in a change of 2.27% for US phone numbers and 0.05% for other phone numbers. The table only includes the top 15 countries by number quantity. The full list is available in the dataset.

Country	Daily median	Min.	Max.	Q1	Q3	Mean daily change (%)
United States	11,751	9,190	15,344	10,812	13,243	266.54 (2.27%)
International	101,913	100,122	102,375	101,711	102,055	54.66 (0.05%)
India	19,446	19,108	19,635	19,118	19,625	6.66 (0.03%)
Indonesia	11,491	11,482	11,502	11,489	11,498	0.54 (0.01%)
Denmark	10,762	10,653	11,358	10,761	11,252	7.03 (0.07%)
Czech Republic	10,038	10,032	10,041	10,036	10,040	0.13 (0.01%)
Poland	9,380	9,287	9,877	9,378	9,661	5.11 (0.05%)
France	8,981	8,956	8,997	8,975	8,986	0.97 (0.01%)
Chile	5,144	5,135	5,155	5,143	5,145	0.47 (0.01%)
Brazil	3,528	3,461	3,618	3,507	3,607	4.51 (0.13%)
Turkey	2,577	2,565	2,595	2,566	2,594	0.33 (0.01%)
Lithuania	1,862	1,851	1,865	1,861	1,863	0.18 (0.01%)
Canada	1,859	1,757	1,919	1,807	1,886	2.65 (0.14%)
Kenya	1,410	1,410	2,410	1,410	2,410	8.45 (0.60%)
United Kingdom	1,715	1,703	1,730	1,713	1,720	0.65 (0.04%)
United Arab Emirates	1,332	1,290	1,353	1,317	1,334	1.17 (0.09%)
Romania	1,314	1,297	1,317	1,300	1,316	0.31 (0.02%)

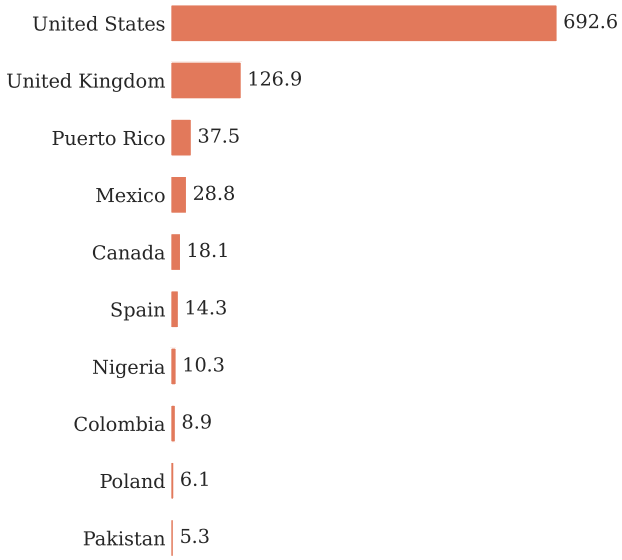


Fig. 4. Per-country statistics: mean number of received calls per available phone number. The international average is 4.25 per available phone number, but some countries such as the United Kingdom or Puerto Rico exhibit a substantial deviation from the mean.

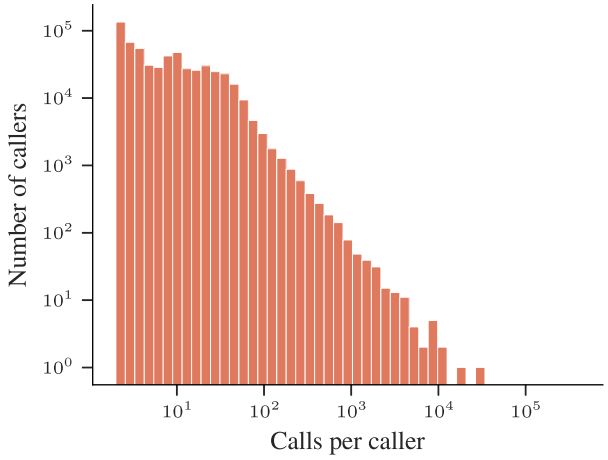


Fig. 5. Distribution of calls per displayed caller ID across the full dataset (US and international data combined) on a log–log scale. Most displayed IDs occur a few times, while a small number account for substantial traffic.

contain observed CDRs. Inventory snapshots are available on 117 dates, and we linearly interpolated the 14 internal gaps. We excluded the 15 dates without CDRs, calls to countries absent from the inventory, calls with `UNKNOWN` callee country, and toll-free calls and inventory that could not be assigned to one country. After these exclusions, the US rate is 3.48 calls per available number-day and the pooled non-US rate is 0.0259. The US-to-non-US rate ratio is 134.41 (95% calendar-week block-bootstrap CI: 106.22–164.47; 18 week blocks). Daily rates are also higher in the US in a paired Wilcoxon signed-rank test (116 shared dates, $W = 1$, $p < 0.001$); the median paired difference is 3.04 calls per number-day with a 95% calendar-week block-bootstrap CI: (1.51, 3.53), with matched-pairs rank-biserial correlation $r_{rb} = 1.000$. Restricting the calculation to the 102 observed inventory dates yields a rate ratio of 128.26.

The above analysis assumed independence per call-level. This assumption could arguably be violated if the calls are disproportionately distributed among the displayed caller IDs, which we analyze and show to be true in our assessment of caller concentration. We propose performing a group-bootstrap approach to remove the limitations of the call-level independence assumption by sampling calls weighted by caller ID, not individual calls, and repeating the process a large number of times (e.g. 10,000). We further refer to this process as the group-bootstrap approach.

In this case, the group-bootstrapped calls per available number-day of US to non-US ratio resulted in a median value of 135.50 (95% CI: (125.80, 146.89)), slightly above the non-bootstrapped version, thereby confirming that the results are persistent across distinct caller IDs.

In Fig. 4, we show the mean number of received calls normalized per phone number availability and additionally split by country. Although the international honeypot mean is 4.25, the UK numbers received 126.9 calls per available number, a substantial deviation from the mean, but still below the US mean of 692.6 calls per available number.

5.1.2. Caller concentration

Fig. 5 shows that calls are unevenly distributed across displayed caller IDs. The US-targeting calls contain 501,960 displayed caller IDs. These IDs appear in a median of 6 calls (IQR: (3, 16)), and the most frequent 1% account for 24.2% of US-targeting calls. The corresponding Gini coefficient is 0.655. The non-US-targeting calls contain 71,982 displayed caller IDs. These IDs appear in a median of 3 calls (IQR: (2, 5)),



Fig. 6. Most frequent non-domestic displayed caller country by destination country. The United States is the dominant external source for 16 destinations, followed by the United Kingdom for 5 destinations.

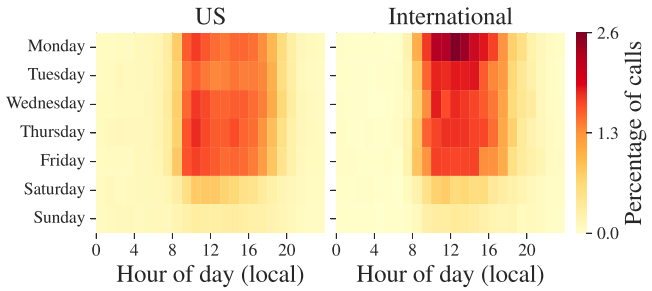


Fig. 7. Temporal distribution of robocalls by hour of day and day of week for US (left) and international (right) callees. Timestamps are in callee local time. Color intensity indicates the percentage of total calls in each region. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and the most frequent 1% account for 23.1% of non-US-targeting calls. The corresponding Gini coefficient is 0.528. These measures describe the concentration of displayed identifiers, not verified callers. Because caller IDs can be spoofed and may belong to uninvolved subscribers, the results neither identify responsible actors nor show that blocking frequently displayed IDs would stop the calls.

We compared caller-ID concentration across the same 18 complete weeks. The weekly Gini coefficient is higher for US-targeting calls by a median of 0.126 (95% week-bootstrap CI: $(0.039, 0.142)$; paired Wilcoxon signed-rank test: $W = 9$, $p < 0.001$, $r_{rb} = 0.895$). Thus, calls are more concentrated among displayed caller IDs in the US data during these weeks. The regional sets share only 119 (0.02%) displayed caller IDs. This small overlap describes the identifiers present in the CDRs, but does not show that the calls originated from separate actors or infrastructure.

The group-bootstrapped analysis of the distribution of calls across displayed caller IDs provided a median Gini coefficient that remained similar for US and non-US targeting calls: 0.656 and 0.528, respectively. The group-bootstrapped weekly Gini coefficient is higher for US-targeting calls by a median of 0.127 (95% CI: (0.096, 0.168)), with the effect being statistically significant ($p < 0.001$).

5.1.3. Temporal patterns

Fig. 7 shows call timing in the callee's local time. Weekday business hours are Monday through Friday from 09:00 to 17:00, and nighttime is 23:00 to 06:00. Weekend and nighttime indicators can overlap and are not parts of one exhaustive composition. In the full corpus, 61.8% of US calls and 69.9% of non-US calls occur during weekday business hours, whereas weekend shares are 12.4% and 10.0%, and nighttime shares are 5.5% and 1.8%, respectively. These patterns are consistent with scheduling, but timing alone does not establish operator intent.

For inference, the unit of analysis is the calendar week. We included the 18 callee-local weeks in which both regions had observations on all seven dates. For each week, we calculated the regional proportions and compared the paired US and non-US values using two-sided Wilcoxon signed-rank tests. The US business-hours proportion is lower by a median of 6.01 percentage points (95% week-bootstrap CI: $(-9.77, -3.33)$; $W = 3$, raw $p < 0.001$, Holm-adjusted $p < 0.001$, $r_{rb} = -0.965$). The US weekend proportion is higher by 0.67 percentage points, although the interval includes zero (95% CI: $(-0.72, 3.06)$; $W = 53$, raw and adjusted $p = 0.167$, $r_{rb} = 0.380$). The US nighttime proportion is higher by 1.83 percentage points (95% CI: $(0.73, 3.79)$; $W = 8$, raw $p < 0.001$, Holm-adjusted $p < 0.001$, $r_{rb} = 0.906$). All differences and rank-biserial correlations are defined as US minus non-US. The confidence intervals were obtained by resampling complete weeks, and Holm correction was applied only to these three temporal comparisons.

We validate the results by performing the previously described group-bootstrap approach. The US business-hours proportion remained lower by a median of 8.14 (95% CI: $(-12.91, -4.98)$), the US weekend proportion remained higher by a median of 2.32 points (95% CI: (0.92, 4.35)), and the US nighttime proportion remained higher by a median of 3.75 points (95% CI: 1.95, 6.69)). The increase in effect sizes obtained using the group-bootstrap approach indicates that the effects are persistent across distinct caller IDs with the weekend proportion becoming statistically significant as the interval excludes zero.

5.1.4. Geographic coverage

The dataset contains 178 displayed caller-country codes and 65 callee countries. US numbers receive 8,314,813 calls (95.1%), followed by Great Britain with 217,787 (2.5%), Poland with 58,356 (0.7%), Canada with 33,373 (0.4%), and Denmark with 31,505 (0.4%).

Caller-country labels were inferred from displayed caller IDs, which can be spoofed and therefore do not establish the caller's location. Overall, 89.9% of calls display a caller country that matches the callee country. This pattern is consistent with neighbor spoofing, but does not prove that spoofing occurred. Across the full dataset, the matching-country proportions are 90.0% for US-targeting calls and 89.2% for non-US-targeting calls. We also compared these proportions across the 18 complete weeks. The weekly matching-country proportion is higher for US-targeting calls by a median of 3.41 percentage points (95% week-bootstrap CI: $(2.16, 5.80)$; paired Wilcoxon signed-rank test: $W = 14$, $p < 0.001$, $r_{rb} = 0.836$). These results measure agreement between displayed caller and callee countries, not the callers' actual locations.

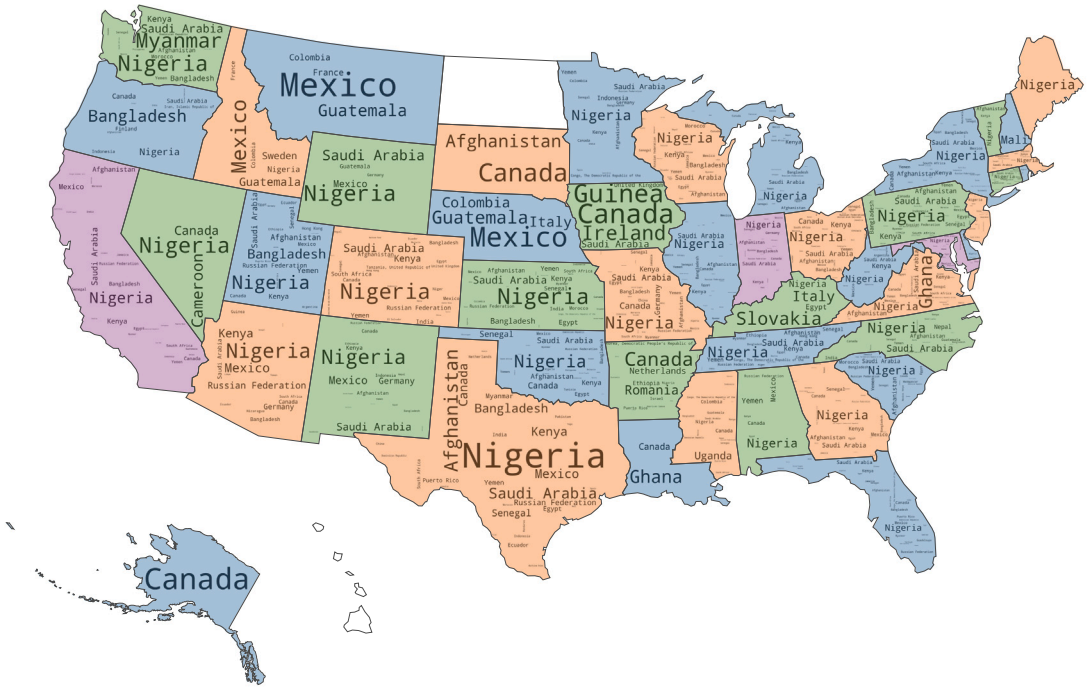


Fig. 8. Word clouds of caller countries by US state, with word size proportional to the number of non-domestic calls. Nigeria is the dominant external source in 35 states, followed by Canada and Mexico, each leading in 4 states.

The group-bootstrap approach provided a median difference of 0.76 points of the weekly matching-country proportion between the US and non-US calls (95% CI: $(-1.08, 2.96)$). In this case, the group-bootstrapped confidence interval encompasses zero, indicating that we cannot confirm the results obtained for the non-bootstrapped approach, i.e. the difference is not statistically significant in this case.

Calls displaying US caller IDs account for 85.7% of all calls, followed by Nigeria (3.9%), Great Britain (2.4%), Saudi Arabia (0.8%), and Canada (0.7%). These labels may reflect spoofing rather than origin. The `UNKNOWN` category (0.6%) denotes unavailable or malformed displayed caller information.

5.2. Regional analysis

We have already noted that domestic traffic makes up the largest share of the dataset, especially for the three leading same-country pairs: the US, the United Kingdom, and Poland. Same-country calls account for 7,866,600 of 8,747,351 total calls (89.9%), with the domestic origin being the single most frequent caller for 54 of 65 destination countries. Given this strong domestic pattern, the next natural step is to examine the most frequent non-domestic callers in the dataset.

Fig. 6 shows the most frequent non-domestic caller country observed in the CDRs for each destination country. After excluding domestic calls, a dominant external caller can still be identified for 47 destination countries, and we are left with 22 unique source (caller) countries. The United States is the most common dominant external source, leading in 16 countries, followed by the United Kingdom in 5 destinations. A small number of other countries, including the Netherlands, India, Slovakia, Germany, Serbia, and Belgium, are the top external caller for two destinations each, while the remaining source countries appear only once.

Here we also see the pronounced volume of calls originating from Nigeria, which is the leading external source for calls to the United States (340,808 of 787,594 cross-border calls). The United States dominates calls to Canada (10,306 of 10,767) and the Philippines (576 of 1,323), the United Kingdom is the dominant source for Denmark (3,553 of 13,253) and Sweden (47 of 126), and India is the top caller for Germany (72 of 73). Overall, the figure shows that international call traffic is not evenly distributed across countries, but is instead concentrated around a relatively small set of recurring country-to-country connections.

We can repeat a similar analysis for traffic to US states. Domestic traffic is again overwhelmingly dominant, as calls from within the United States account for 7,375,854 of 8,143,011 calls to US states with a resolvable caller country (90.6%), and the domestic source is the largest contributor in all 51 covered jurisdictions (50 states and the District of Columbia). For this reason, it again makes sense to focus the analysis on non-domestic calls.

Fig. 8 shows the distribution of non-domestic caller countries in the CDRs for each US state, with word size indicating frequency and the largest word corresponding to the dominant external source. Even after excluding domestic calls, the remaining external traffic is still fairly concentrated. California has the largest absolute number of non-domestic calls (123,197, 16.1% of non-domestic calls), followed by New York (66,254, 8.6%) and Massachusetts (66,212, 8.6%), while Nigeria is the dominant external source in the largest number of states, leading in 35. Canada and Mexico each lead in four states, Ghana leads in two, and Bangladesh, Mali, Slovakia, and Uganda each lead in one state. Note that there were no non-domestic calls recorded for North Dakota and Hawaii. Overall, the figure shows that non-domestic call traffic to the United States is concentrated among a relatively small set of external origins rather than being evenly distributed across states.

5.3. Temporal co-targeting and transcript similarity

Displayed caller IDs are observable in the CDRs, but they are not reliable actor identifiers because they can be spoofed. We therefore test whether IDs grouped based on shared targets and call timing also exhibit similar call content. First, we connect IDs that target the same honeypot numbers within a defined time window. Second, we partition the resulting graph into groups. Third, we compare transcript similarity within and between these groups. The groups are constructed using

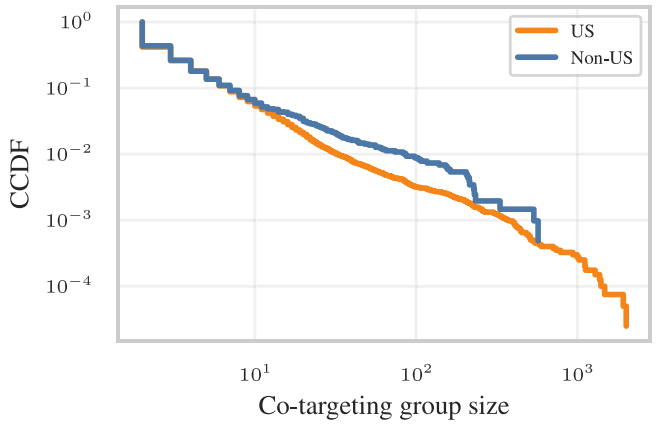


Fig. 9. Complementary CDF of temporal co-targeting group sizes for US and non-US callees. Both distributions are heavy-tailed: a large majority of groups contain only a few displayed caller IDs, while a small number of groups exceed 10^2 members. Groups of size 1 are excluded.

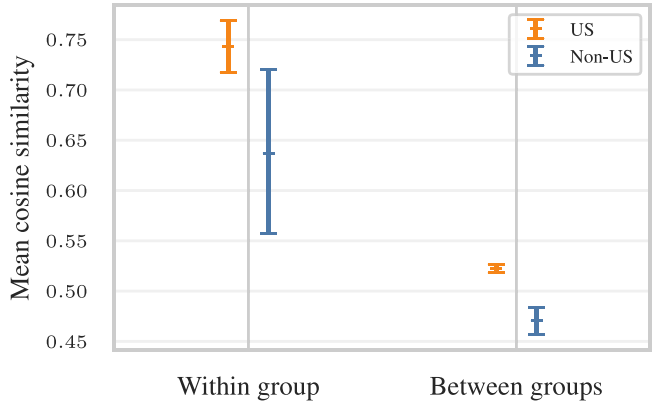


Fig. 10. Group-level cosine similarity of caller-ID-averaged BGE-M3 transcript embeddings within and between temporal co-targeting groups for US and non-US callees. Markers show means across eligible groups, with each group receiving equal weight; error bars are percentile 95% bootstrap intervals obtained by resampling groups.

call metadata only, while the transcripts are used to check whether the same structure is also visible in call content.

For this analysis, we apply the preprocessing described above and retain displayed caller IDs observed at least twice. Unlike the preceding metadata analysis, we include records from both passive and interfering modes because interfering-mode calls are needed to link graph nodes to call transcripts. Regions are assigned using the known callee country and reported as US and non-US. A target called by many different IDs can connect IDs that do not otherwise show related behavior in the projected graph (Tumminello et al., 2011; Neal, 2014). We therefore exclude the ten targets called by the largest numbers of distinct displayed caller IDs in each region. We test whether this choice affects the results by repeating the analysis while excluding 0, 5, 10, and 20 targets.

The intuition behind the graph construction is that displayed caller IDs involved in the same campaign may dial the same honeypot numbers at around the same time. We build an undirected graph separately for US and non-US callees, in which each node is a displayed caller ID. We connect two IDs when they call the same honeypot number within a 24-hour window and weight the connection by the number of targets they share. We then group the connected IDs using weighted Leiden community detection (Traag et al., 2019). The resulting US graph contains 201,913 nodes and 702,984 edges, while the non-US graph contains 12,255 nodes and 33,812 edges. Most groups correspond to complete connected components (99.7% in the US graph and 99.1% in the non-US graph), while Leiden divides a small number of larger components into smaller groups. The resulting group-size distribution (Fig. 9) is heavy-tailed in both regions: most groups contain only a few IDs, while a small number of groups are large.

Having grouped the displayed caller IDs using metadata alone, we next ask whether the same groups are reflected in what the calls say. The transcript inputs were previously limited to recordings containing at least 25% detected speech, and we additionally retain cleaned transcripts containing at least 40 characters. This analysis uses the larger collection of automatically generated transcripts, rather than only the 839 human-verified transcripts released publicly. We use the CDRs to match transcripts to displayed caller IDs.

We represent each transcript using BGE-M3 (Chen et al., 2024), a multilingual embedding model that places texts with similar meanings close together. For caller IDs with multiple transcripts, we average their embeddings to obtain one vector per displayed caller ID. Of the 201,913 US graph nodes, 1,366 (0.68%) have a usable transcript; the corresponding count is 109 of 12,255 non-US nodes (0.89%). Within-group comparison requires at least two transcript-covered IDs in a group. This leaves 786 US IDs across 207 groups and 80 non-US IDs across 19 groups, producing 2,526 and 215 within-group pairs.

The test is straightforward. If co-targeting groups capture common calling behavior, displayed caller IDs inside the same group should have more similar call content than IDs from different groups. For each group, we compute the mean cosine similarity over all pairs of transcript-covered IDs in that group. This includes all within-group pairs, whether or not the two IDs share a direct graph edge. Formally, for a group g with caller-level vectors $\{\mathbf{v}_i\}$,

$$\bar{s}_{\text{within}}(g) = \frac{1}{\binom{|g|}{2}} \sum_{i<j} \cos(\mathbf{v}_i, \mathbf{v}_j), \qquad \cos(\mathbf{v}_i, \mathbf{v}_j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \, \|\mathbf{v}_j\|}. \tag{1}$$

For the between-group comparison, we calculate the mean cosine similarity between the IDs in each group and all transcript-covered graph IDs outside that group but within the same region. We then calculate the difference between the within- and between-group means:

$$D_g = \bar{s}_{\text{within}}(g) - \bar{s}_{\text{between}}(g). \tag{2}$$

We average these differences across groups, giving each group equal weight. Confidence intervals are obtained by resampling groups with replacement over 2,000 bootstrap iterations and recomputing the mean difference. The confidence intervals are reported separately for each region and are not adjusted for multiple comparisons.

For US groups, the mean within-group similarity is 0.743, compared with a between-group mean of 0.523. The mean difference is 0.220 (95% group-bootstrap CI: $[0.196, 0.245]$). For non-US groups, the corresponding means are 0.637 and 0.471, giving a mean difference of 0.167 (95% CI: $[0.092, 0.247]$). Within-group transcript similarity is higher on average among eligible groups in both regions.

Both intervals exclude zero, indicating that within-group transcript similarity is significantly higher than between-group similarity. Fig. 10 shows the group-averaged within- and between-group similarities and their bootstrap intervals.

The transcript comparison shows that displayed caller IDs grouped using call timing and shared targets also have more similar call content. This content coherence is consistent with campaign-like organization, but it does not establish that every group corresponds to one campaign or actor. Displayed caller IDs can be spoofed, and transcript coverage is incomplete. Recording and transcript filters may favor some types of calls, so the observed similarity differences may not represent groups with few or no usable transcripts.

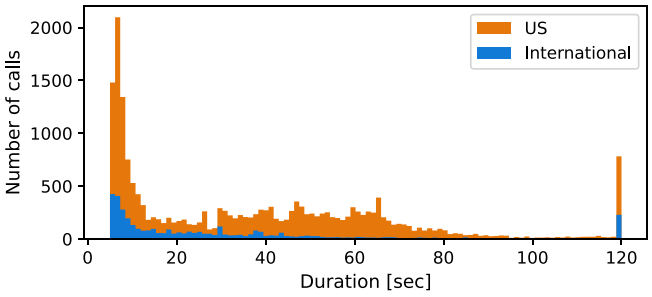


Fig. 11. Duration of recorded US and international phone calls. The distributions for both regions follow a similar pattern, with most calls being shorter than 20 s.

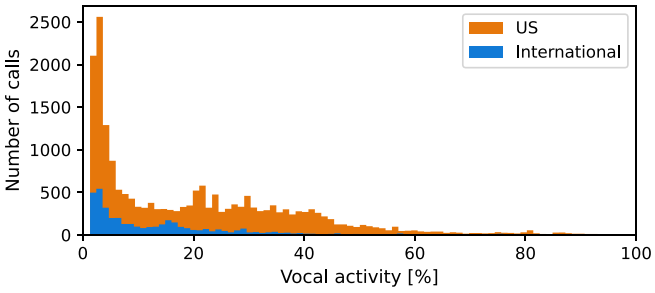


Fig. 12. Percentage of vocal activity detected. In both regions, a substantial amount of robocalls are silent, with very little vocal activity. This is a common phenomenon among robocalls and has been previously observed (Prasad et al., 2020, 2023). "Blank" calls are usually made with the goal of determining whether the dialed number has voice capabilities (scouting calls).

6. Call content analysis

In this section, we analyze the audio content of the recorded calls. In total, we have recorded 263,099 calls while the honeypot was operating in its interfering mode, of which 220,301 calls were made to the US and 42,798 calls to the rest of the world. Due to the aforementioned legal reasons, calls shorter than 5 s were discarded, after which 197,765 (89.77%) US calls were left and 33,253 (77.7%) international calls remained. We can therefore only speculate about the nature of the shorter calls, as we have no way of discerning what percentage of those calls belong to robocalls or to legitimate, human calls. A histogram depicting the durations of the recorded calls is shown in Fig. 11.

We assess whether there is a difference between the distributions of call durations between the US and international calls. Given that our data is not normally distributed, we use the non-parametric Mann–Whitney U test (Mann and Whitney, 1947). The results indicated that the duration of calls made in the US ($M_d = 32.97$ s IQR : $(8.66, 57.92)$) were significantly higher than the duration of international calls ($M_d = 19.52$ s, IQR : $(7.94, 41.49)$), $U = 35{,}264{,}761$, $p \ll .001$. The rank-biserial correlation indicated a small effect size ($r_{rb} = .15$).

The Mann–Whitney U test above assumed that each call can be considered an independent observation, an assumption that we consider violated to some extent, given that a large number of calls can be attributed to the same callers. We therefore implement a group-bootstrapping approach, similar to the one used in the metadata analyses and the co-targeting analysis. We sample a new dataset with the same size as our original dataset, but instead of attributing equal weights to every sample (call), we attribute equal weights to each caller. The entire process is repeated 10,000 times, from which we average the results and report on the confidence intervals.

With the group-bootstrapping approach we still obtain a statistically significant result ($U = 37{,}901{,}717$, 95% CI: (37,876,658, 37,926,514), $p \ll .001$), but the rank-biserial correlation is lower: $r_{rb} = .084$ (95% CI: (.083, .085)), indicating a lower effect size, which is expected.

6.1. Voice activity detection

Previous work has shown that a notable percentage of unsolicited calls are silent, or contain very little spoken content. In the study by Prasad et al. (2020), 62.75% of the calls had less than 10% spoken content, while 42.05% had less than 1% of audio content, while Prasad et al. (2023) report that 31.28% of calls over a collection of more than 1.3 million were completely silent. Prasad et al. (2025) only analyze calls that contain at least 5 s of audio and 10% of audio content, with these thresholds being chosen based on estimates of how long a typical English speaker takes to deliver a meaningful short sentence in the context of a robocall campaign.

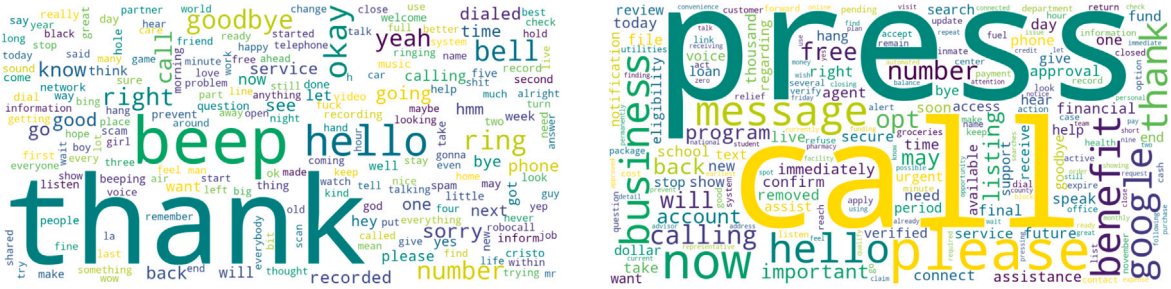
Fig. 12 shows the distribution of the percentage of spoken content within US and international calls. We used SileroVAD (Silero Team, 2024), an industry-grade voice activity detection (VAD) module designed specifically for use in telephony, to detect vocal segments of audio. With SileroVAD, we observed that a total of 50,911 (24.78%) calls contain any vocal activity at all in the US, compared to 13,639 (41.02%) internationally. The relative difference between the proportions of calls with vocal activity in the US and international calls is 65.54%, indicating that calls in the US are more likely to be silent. However, for voiced calls we report that US calls and international calls are not too different. Although we found a statistically significant difference in the proportion of vocal activity between US and international calls for non-silent calls using a Mann–Whitney U test ($U = 37{,}347{,}940$, $p \ll .001$), the effect size was negligible to small ($r_{rb} = .097$), indicating that voiced calls contain a similar amount of vocal content in both the US and international settings.

By group-bootstrapping the calls the statistical significance remains ($U = 38{,}539{,}176$, 95% CI: (38,512,236, 38,565,639)), but the effect size is almost negligible: $r_{rb} = .069$, 95% CI:(.068, .069), thereby reaffirming that voiced calls contain a similar amount of vocal content in the US and international settings.

6.2. Transcript analysis

Having determined the amount of spoken content, we proceed to transcribe the calls. We transcribed all calls using Whisper (Radford et al., 2023), a state-of-the-art open-source automatic speech recognition model (specifically model version *large-v3-turbo*). Our subsequent analysis is based on a heuristically determined threshold which is that the calls should contain at least 25% spoken content, in line with previous work (Prasad et al., 2020). The reason for this is that transcribing such a large number of calls is done automatically, and state-of-the-art transcription models suffer from hallucinations when dealing with audio that has a low percentage of spoken content (Koenecke et al., 2024; Barański et al., 2025).

We justify our decision and illustrate this by separately transcribing US recordings that contain less than 25% of spoken content (low speech ratio) and recordings that contain more than 25% of spoken content (high speech ratio). We extract the most commonly occurring words from both sets of recordings and represent them as word clouds in Fig. 13. By far the most commonly occurring word in the low speech ratio recordings is the word *thank*, which among others is one of the most frequent and well-known hallucinations produced by Whisper: *"thank you"* (Koenecke et al., 2024; Barański et al., 2025). The word *beep* is the second most spoken word in the low speech ratio recordings, while it does not even appear in the word cloud of the most common words for the high speech ratio recordings. The words *thank* and *hello* are fairly common in both sets of recordings, but the differences are clear. Action verbs such as *call* and *press* top the high speech ratio recordings,



(a) Wordcloud for low speech ratio recordings.

(b) Worldcloud for high speech ratio recordings.

Fig. 13. Hallucinations caused by transformer-based automatic speech recognition can heavily impact the transcriptions. Hallucinations are more prominent in recordings that contain a lower ratio of speech compared to silence. To the left, we show the most common words occurring in low speech ratio recordings, while the image on the right shows the most frequently occurring words in high speech ratio recordings. There is a clear difference between the distribution of words between the two groups, and the words on the left correspond to well-known and frequent hallucinations produced by Whisper (Koenecke et al., 2024; Barański et al., 2025).

Table 2
Most commonly occurring languages.

US	International
English (92.35%)	English (59.44%)
Spanish (3.80%)	Spanish (13.85%)
Mandarin Chinese (1.30%)	Polish (8.23%)
Korean (0.71%)	Hindi (7.10%)
French (0.43%)	Czech (1.82%)

followed by words that indicate a business matter or sense of urgency: *important*, *message*, *account*, *business*, *google*, *service*, *number*, etc. We therefore limit further analysis to recordings that contain at least 25% spoken content, leaving 20,328 US robocalls and 4,071 international robocalls.

6.2.1. Most common languages

In Table 2 we provide the top 5 most commonly occurring languages and the percentage of their representation within the recordings for the US and international calls. The language detection is provided by Whisper (Radford et al., 2023), which in addition to the transcription also detects the language of the recording automatically. As expected, English is the most frequently occurring language; in the US, it accounts for 92.35% of the calls, an overwhelming majority, followed by Spanish (3.80%) and Mandarin Chinese (1.30%). In international calls, English is still the most common language (59.44%), but other languages occur more frequently than in US calls.

6.2.2. Similarities across different languages

Table 3 shows the top 10 most frequent words occurring in international calls for English, Spanish, and Polish with their translations into English and the normalized word frequency:

$$f_{\text{norm}}(w) = \frac{\text{count}(w)}{\text{count}(w_{\max})} \tag{3}$$

where $\text{count}(w)$ represents the total number of occurrences of a given word w, and $w_{\max}$ represents the most commonly occurring word. Although not identical, the most frequent words for all three languages are words that can be associated with the context of telephone conversations (*press* and *number*). Words that could be attributed to robocall activity such as *loan*, *payment*, *credit*, *important*, *approved*, and *rate* are present in all three languages.

6.2.3. Regional differences between the same language

We want to inspect whether there are differences between the most commonly occurring words of the same language based on the region, i.e. US or international. To illustrate this, we show the word clouds for English and Spanish, per US and international data, in Fig. 14. Although the exact order and ranking of the words do differ per region, for English the most frequently occurring words are similar in the US and the international recordings, with *press* being the first and second most frequently used word. Words commonly associated with robocalls such as: *loan*, *offer*, *payment*, *now*, *important*, and *financial*, are present in both regions.

One of our observations was that the word *call* is the joint-top occurring word in US calls, but is below the top 10 words in international calls, while the word *press* is the top word in both regions. This pattern is present in Spanish recordings as well. The word *llamada* meaning *call* is the most frequent word in US recordings, but is not one of the most common words in international recordings. This may be an indication that robocallers are more likely to use similar scams in the same region, but in different languages, rather than similar scams in the same language but in different regions.

Similarly, the words *oprima* and *presione*, which can both be translated as *press*, are the most frequent words in international recordings. This may indicate that in the US, robocallers are more likely to try to trick the target into calling a specific number, while in international robocalls the goal is to make the victim interact with the robocall. These claims would have to be verified with further analysis.

6.2.4. Relationship between language and callee country

The analysis so far was limited to analyzing the content separately of the caller and callee information. Including the caller number in the analysis can provide some information, but would be of limited reliability since it could be spoofed. We are, however, certain about the callee number, since it belongs to our honeypot. In Fig. 15, we present a chord diagram, showing how the recordings, based on language, are distributed among destination countries. An expected result is that English is the most widespread language, with robocalls in English targeting numbers in a majority of countries. Robocalls in Spanish are internationally distributed as well, with calls in Spanish directed towards Bahrain, Chile, India, Mexico, the Philippines, and Poland. Calls in Hindi are directed towards countries in the Indian subcontinent: India and Pakistan. Polish calls are directed overwhelmingly towards Poland, with a small number directed towards the United Kingdom, Romania, and Ukraine, the latter two of which are Slavic countries like Poland. Overall, the language of the robocalls is aligned with native languages of the destination countries, while robocalls in English are globally distributed and English speaking countries are targeted globally.

6.3. Identifying robocalls and campaigns

In this subsection, we identify robocall campaigns and classify the most common types of robocall scams.

Table 3
Most frequent words among English, Spanish, and Polish international recordings.

English	Spanish	Polish
Press (1.00)	Oprima (press) (1.00)	Sie (myself) (1.00)
Pay (0.58)	Presione (press) (0.55)	Numer (number) (1.00)
Loan (0.42)	Bueno (good) (0.43)	Prosze (please) (0.47)
Dear (0.34)	Malo (bad) (0.38)	Dzien (day) (0.46)
Now (0.31)	Sabia (know) (0.32)	Dobry (good) (0.45)
Customer (0.26)	Informacion (information) (0.29)	Zostalo (remaining) (0.33)
Legal (0.24)	Credito (credit) (0.28)	Online (online) (0.32)
Offer (0.23)	Ejecutivo (executive) (0.26)	Zatwierdzone (approved) (0.32)
Payment (0.21)	HSBC (HSBC) (0.25)	WhatsApp (WhatsApp) (0.32)
Call (0.20)	Importante (0.24)	Stawka (rate) (0.32)



(a) US recordings, English language.

(b) US recordings, Spanish language.

(c) International recordings, English language.

(d) International recordings, Spanish language.

Fig. 14. Word clouds for English and Spanish, generated separately for US recordings and international recordings. Although robocalls were made in the same language, the word clouds differ, indicating that language is not the only predictor of robocall content — the region matters as well.

6.3.1. Campaign discovery via transcript clustering

Similar to Prasad et al. (2025), we apply Density-Based Spatial Clustering of Applications with Noise (DBSCAN) (Ester et al., 1996) to identify robocall campaigns based on their transcripts.

Text normalization. Each transcript undergoes a text normalization process in which we mask sensitive and variable information that should not impact the assignment of a transcript to a given cluster. For example, telephone numbers are replaced with the string "[phone]", monetary amounts with "[amount]", URLs with "[url]", and personal names with "[name]".

Feature representation. Normalized transcripts are represented as TF-IDF vectors over the vocabulary of unigrams and bigrams. We use sublinear term frequency scaling replacing the raw linear term count with a logarithmic scale:

$$\mathrm{tf}(t,d) = 1 + \log(\mathrm{count}(t,d)) \tag{4}$$

Terms appearing in fewer than 10 transcripts (documents) or in more than 90% of transcripts are ignored. Transcripts shorter than 40 characters are skipped as well. The vocabulary is capped at 15,000 features. These values were determined empirically after manually assessing the quality of the clusters by comparing the similarity of the transcripts within each cluster. The resulting sparse vectors are $L2$ normalized, so that the cosine similarity between two transcripts $\mathbf{u}$ and $\mathbf{v}$ is equal to the dot product between them:

$$\cos(\mathbf{u},\mathbf{v}) = \mathbf{u}\cdot\mathbf{v} \tag{5}$$

Clustering. We use DBSCAN (Ester et al., 1996) to cluster the processed transcripts, represented in the TF-IDF feature space, in an unsupervised manner. DBSCAN has several benefits: (i) it does not require us to specify the number of clusters a priori, (ii) it inherently handles noise and can ignore transcripts that do not belong to any cluster, and (iii) it is agnostic to cluster shape and size. DBSCAN has two parameters which we varied: $\texttt{minPts} \in \{10, 50, 100\}$ and $\epsilon \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ by manually inspecting how well the transcripts were clustered. We finally used the configuration $\epsilon = 0.4$ for both regions, while `minPts` was 50 for US transcripts and 10 for international transcripts.

We apply the clustering procedure only to transcripts that have more than 25% speech content. Combined with filtering transcripts based on their length, this left 16,177 US transcripts and 2,701 international transcripts for clustering. A majority of the transcripts are considered to be noise, 11,693 in the US and 1,894 internationally. In Tables 4 and 5 we show the top 5 clusters of each region.

Privacy and availability. We make publicly available a total of 724 US transcripts and 115 international transcripts which have been verified not to contain any personal information or have been anonymized to conceal first names.

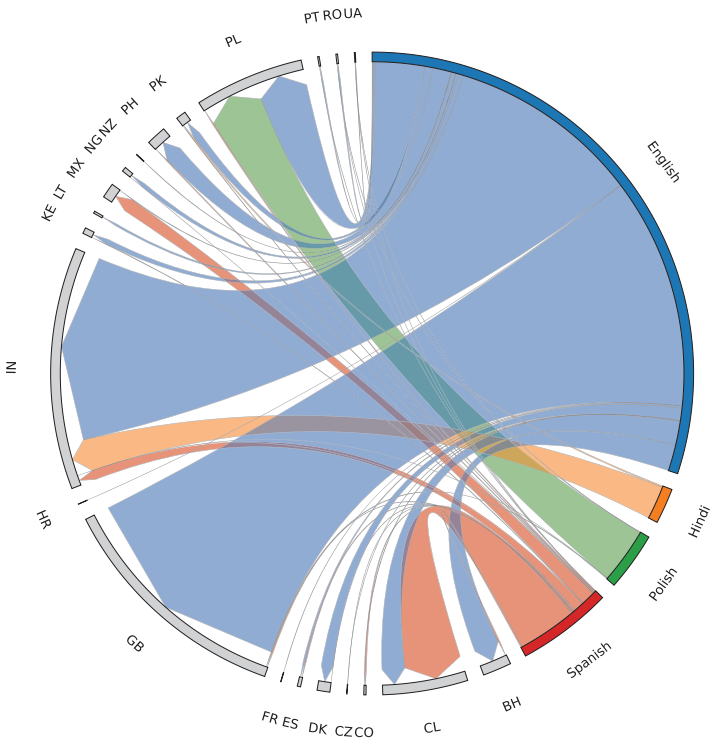


Fig. 15. Chord diagram showing the flow of languages towards callee countries. Country names are in ISO 3166-1-alpha-2 codes.

Table 4
Top 5 discovered US robocall clusters. The most common type of robocalls are tech scams, most notably impersonations of the company Google.

Cluster	Count	Top words
Benefits/Grants scam	460 (10.26%)	Benefits, period, program, now, press, case
Google business listing scam	245 (5.46%)	Google, voice, business, press, search, customers
Google business listing scam (II)	239 (5.33%)	Google, agent, voice, business
Exclusive benefits spam	204 (4.55%)	Benefits, exclusive, thousands, quickly, pocket
Caller ID spoofing	196 (4.37%)	Return, assist, urgent, time-sensitive, capital recovery

Table 5
Top 5 international robocall clusters. The most common scam types are related to loan/debt scams, with 3 out of the top 5 clusters belonging to this type of scam. Scams are multilingual: English, Hindi, and Spanish.

Cluster	Count	Top words
Loan/Debt scam (Hindi)	83 (10.29%)	Legal, recovery, notice, final, agent
Loan/Debt scam (II) (English)	59 (7.31%)	Qualified, loan, prequalified, Whatsapp, link
PayMe loan (English)	43 (5.33%)	PayMe, loan, reminder, visit, delay
Political spam (Spanish)	40 (4.96%)	Important, mayor, performance, opinion
Loan/Debt scam (III) (Hindi)	35 (4.34%)	Loan, overdue, legal, notice, visit

6.3.2. *Few-shot robocall categorization*

Large language models (LLMs) have recently emerged as a popular way of processing textual data due to their ability to perform downstream tasks without additional fine-tuning, in a zero-shot or few-shot manner (Brown et al., 2020; Wei et al., 2022; Sun et al., 2023; Mu et al., 2024). LLMs can be instructed and guided to perform tasks such as classifying audio transcripts as robocalls and categorizing them. We perform few-shot classification: the LLM is provided with a sufficiently detailed description of the problem (classifying robocalls) in its system prompt, and the description contains a few examples of (non)-robocalls. We used ChatGPT-5.4 to classify robocalls based on their transcript.

The LLM categorized 11,148 (54.8%) of US robocalls and 3,021 (74.45%) of international robocalls. It could classify robocalls into eight categories: Business and Marketing, Financial Services, Government Impersonation, Health and Insurance, Legitimate Automated Notification, Political and Non-Profit, Technical Support Scam, and Unknown/Other. The exact distribution of categories per US and international groups is shown in Fig. 16. Table 6 shows a typical robocall from each category. The results of using an LLM to directly classify robocalls are aligned with the robocall clustering approach: both approaches showed that tech support scams are more common in the US, with financial services scams being the most dominant type of scam internationally.

We performed a chi-square test of homogeneity to assess whether there is a significant difference between the categories of robocalls across regions and obtained a positive result ($\chi^2(7) = 617.23,\ p \ll .001$). The null hypothesis is that the distribution of robocall categories is identical across regions, i.e. there is no difference between the distribution of robocall categories between US and international robocalls. We used post-hoc analyses using adjusted standardized residuals (Haberman, 1973), for which the value of the adjusted standardized residual (Z) indicates the deviation of the data distribution per category (expressed in units of standard deviation) compared to the null hypothesis. The results showed that there is a statistically significant difference for most categories; *Financial Services* are a significantly more common

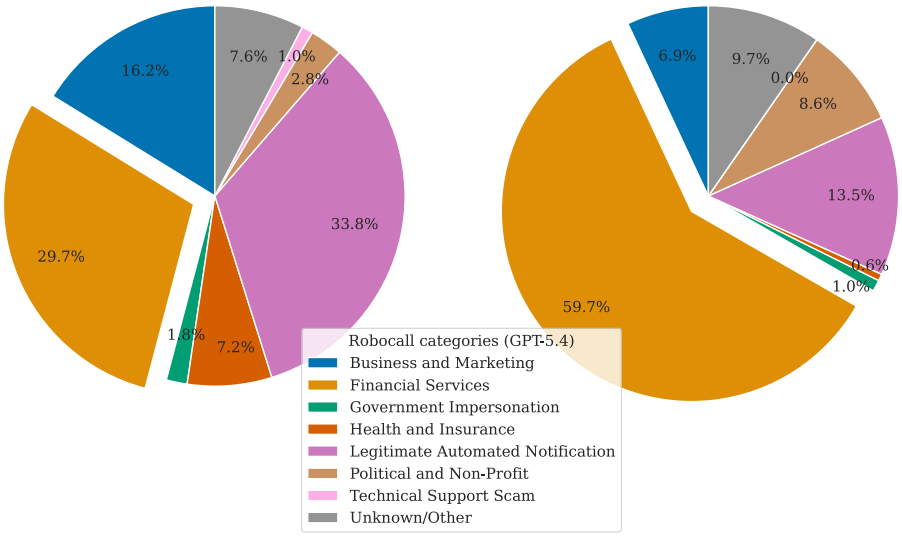


Fig. 16. Robocall categorization per region (left – US, right – international). Similar to the results obtained with robocall clustering (Tables 4 and 5), scams related to financial services are more common in international robocalls, while tech scams are more common in US robocalls. Additionally, US robocalls contain a larger percentage of legitimate automated notifications.

Table 6
Extracted robocall categories and examples of typical robocall scripts from each category. A quick internet search on the callback numbers from the transcripts reveal that these numbers are well-known robocallers with bad caller reputation.

Category	Example
Business and Marketing	Hello, your timeshare details just came across my desk and there is something you should know. Call me back ASAP 949-237-7769. Again, that phone number is 949-237-7769. Thank you. Talk to you soon.
Financial Services	Hello, this is Personal Loan Locator, just giving me a real quick call in regards to the recent personal loan request you submitted. I have a couple alternative options for you, as well as some additional resources to go over. So go ahead and give me a call back at 855-926-6692. Again, that phone number was 855-926-6692. To opt out, go ahead and press 9. Thank you.
Government Impersonation	This is Victoria Masterson with Pre-Legal Services. This is our final attempt to notify you regarding legal documents that have been returned. Please contact my office at 301-298-1127. Again, that's 301-298-1127. Thank you.
Health and Insurance	Eligibility Patient Advocacy Liaisons is calling Samia to provide a free service regarding a personal matter. This is a free service. This is not a sales or collections call. Please call ePALS back at 346-738-0894 and quote reference number account number. Again, our number is 346-738-0894 and quote reference number account number. Thank you. Enlaces de apoyo para el paciente lo es de la mando a Samia para ofrecer un servicio gratuito con respecto a un asunto personal, no es una la manda de ventas o cabranzas. Favor de la mando enlaces de apoyo para el paciente al telefono 346-738-0894 con el número de referencia account number. De nuevo, el número de teléfono es 346-738-0894 y el número de referencia es account number. Gracias.
Legitimate Automated Notification	This message is in English and Espanol. This is a message from the Department of Public Works. Please note there is a household hazardous waste drop-off day this Saturday, October 25th from 9 a.m. to 12 p.m. at the Wastewater Treatment Plant, 40 South Porter Street. Proof of Haverhill Residency is required, and there is a 25-pound-slash-25-gallon limit per vehicle. Este es un mensaje del Departamento de Obras Públicas. Tenga en cuenta que hay un día de entrega de desechos domésticos peligrosos este sábado 25 de octubre de 9 a.m. a 12 p.m. en la planta de tratamiento de aguas residuales, 40 South Porter Street. Se requiere prueba de residencia en Averillie y hay un límite de 25 libras, barra 25 galones por vehículos.
Political and Non-Profit	Greetings, neighbors. This is your 16th Ward Alderman Stephanie Coleman, and I warmly invite you and your family to the 16th Ward monthly community meeting and turkey giveaway on Saturday, November 8th. Join us at a special location for this event only, the Salvation Army Red Shield Adele Center, located 949 West 69th Street from 10 a.m. until 12 noon. We will have representatives from the Chicago Department of Public Health, Howard Brown Health, and the Greater Chicago Food Depository to share valuable information and resources available for our community, along with a light holiday meal to enjoy together. Be a part of the change that's happening in the 16th Ward as we build stronger communities together. Mark your calendars for Saturday, November 8th at a special location, Salvation Army Red Shield Adele Center at 949 West 69th Street from 10 a.m. until 12 noon. Don't just meet me there, beat me there.
Technical Support Scam	This call is inform you that your Amazon Prime subscription has been auto-renewed and $499 has been debited from your account successfully. If you want to cancel this subscription, please press 1 to talk to our executive.

type of robocall category in international calls compared to US calls ($Z = +19.81$). Similarly, category *Political and Non-Profit* ($Z = +9.97$) is disproportionally more present in international calls compared to US calls. Categories *Business and Marketing* ($Z = -7.89$), *Health and Insurance* ($Z = -8.17$), and *Technical Support Scam* ($Z = -3.17$) are more common in US robocalls compared to international calls, while the biggest disproportion favoring US robocalls compared to international robocalls was observed in *Legitimate Automated Notifications* ($Z = -13.38$). The disproportion observed for category *Government Impersonation* ($Z = -1.93$) is not statistically significant. We adjusted for multiple comparisons using the Bonferroni correction.

The chi-square test was repeated using the group-bootstrap approach, previously described in the beginning of the section. The overall result remained statistically significant ($\chi^2(7) = 431.37$, 95% CI: (432.58, 437.81), $p \ll .001$), but the adjusted standardized residuals have changed. The category *Government Impersonation* (−2.05) still is the only category where the differences between the US and international robocalls are not statistically significant. *Financial Services* (+15.12) remains the category with the most prominent difference

between US and international robocalls, but the magnitude has slightly decreased. On the contrary, category *Political and Non-Profit* (+10.32) now exhibits the second largest difference. The ordering of the remaining categories has changed, but remains statistically significant: *Business and Marketing* (−9.18), *Health and Insurance* (−8.01), *Legitimate Automated Notification* (−7.44), and *Technical Support Scam* (−3.70).

6.3.3. Validating the annotation labels

Robocalls were categorized using zero-shot prompting, but we have no ground truth to compare the categorization against. To estimate the reliability of the LLM categorization we drew a stratified random sample of 400 robocalls (50 per category) and had the samples categorized by another LLM model, Gemini-3.1 Pro, and a human annotator (HA).

We first obtained Fleiss' $\kappa = 0.698$, indicating an inter-annotator agreement between the two LLMs and the human annotator between moderate-to-good and substantial (Landis and Koch, 1977). Post-hoc pairwise agreement rates were measured using Cohen's κ: (GPT-5.4 and Gemini-3.1 Pro: $\kappa = 0.799$), (GPT-5.4 and HA: $\kappa = 0.635$), and (Gemini-3.1 Pro and HA: $\kappa = 0.660$).

The disaggregated, per-category Fleiss' κ shows that the inter-annotator agreement varies substantially depending on the robocall category: (Unknown/Other: 0.3), (Legitimate Automated Notification: 0.585), (Government Impersonation: 0.678), (Health and Insurance: 0.686), (Financial Services: 0.727), (Business and Marketing: 0.744), (Political and Non-Profit: 0.858), and (Technical Support Scam: 0.875). Most disagreements can be attributed to the category Unknown/Other and are caused by the HA being less likely to assign robocalls to this category, while, on the other hand, the LLMs were more likely to categorize robocalls as Legitimate Automated Notifications compared to the HA. Excluding these two categories results in an overall Fleiss' $\kappa = 0.865$, with pairwise Cohen's κ: (GPT-5.4 and Gemini-3.1 Pro: $\kappa = 0.970$), (GPT-5.4 and HA: $\kappa = 0.818$), and (Gemini-3.1 Pro and HA: $\kappa = 0.807$), a substantial improvement.

We conclude that the data suggests a higher agreement rate between LLM-LLM categorization compared to human-LLM agreement. This is not evidence that the labels assigned by the LLMs are more correct; LLM training biases could inflate the LLM to LLM agreement, thus the numbers should be anchored to the human annotator agreement. Regardless of the disproportionate LLM to LLM agreement against LLM to human agreement, a substantial level of agreement was observed for 6/8 categories (Landis and Koch, 1977). Additional research and experiments involving more human annotators could provide more concrete results but exceed the scope of this paper.

6.4. Analyzing audio quality

We analyzed the quality of audio recordings using DNSMOS P.835 (Deep Noise Suppression Mean Opinion Score) (Reddy et al., 2022), a non-intrusive audio quality predictor that is based on the subjective test defined by the ITU-T P.835 (International Telecommunications Union Telecommunication Standardization Sector, 2003). DNSMOS P.835 provides three scalar estimates of the quality of an audio clip: SIG (speech), BAK (background noise), and OVRL (overall audio quality). The scores range from 1 (very poor) to 5 (excellent). The Pearson Correlation Coefficient (PCC) between SIG, BAK, and OVRL estimates and human ratings is 0.94, 0.98, and 0.98, respectively, thereby indicating a high degree of correlation to the subjective perception of audio quality and the model's prediction (Reddy et al., 2022).

6.4.1. Impact of audio quality on transcription accuracy

Automatic speech recognition models are not perfect. Ideally, it would be good to manually transcribe the dataset, but this is difficult to achieve given the large volume of the dataset and its multilinguality. We do report, however, that annotators commented on no substantial deviations of the spoken content compared to the automatically transcribed content and were overall satisfied with the generated transcriptions.

Audio quality is correlated with the accuracy of WER. Dubey et al. (2024) showed that the SIG component of DNSMOS P.835 correlates the most (PCC = 0.647) with the word accuracy (WAcc) of automatic speech recognition systems. The formula defining WAcc is as follows

$$\mathrm{WAcc} = 1 - \mathrm{WER} \tag{6}$$

where WER represents the word error rate of the ASR system compared to transcribed speech. The correlation between BAK and OVRL with WAcc was found to be PCC = 0.150 and PCC = 0.526, respectively. Gallardo et al. (2017) showed that the expected WER of an ASR system can be estimated from the quality rating of the audio, for example, transcoding audio using G.711 (International Telecommunication Union Telecommunication Standardization Sector, 1988), the international standard for telephony, resulted in a WER increase by 4.8%. Given Whisper's standard benchmark reports a WER of 2.01 (Radford et al., 2023; OpenAI, 2023) on the clean LibriSpeech dataset, we can coarsely estimate the WER to about 6.81% since we are dealing with G.711 transcoded audio.

6.4.2. Correlation between audio quality and robocall classification

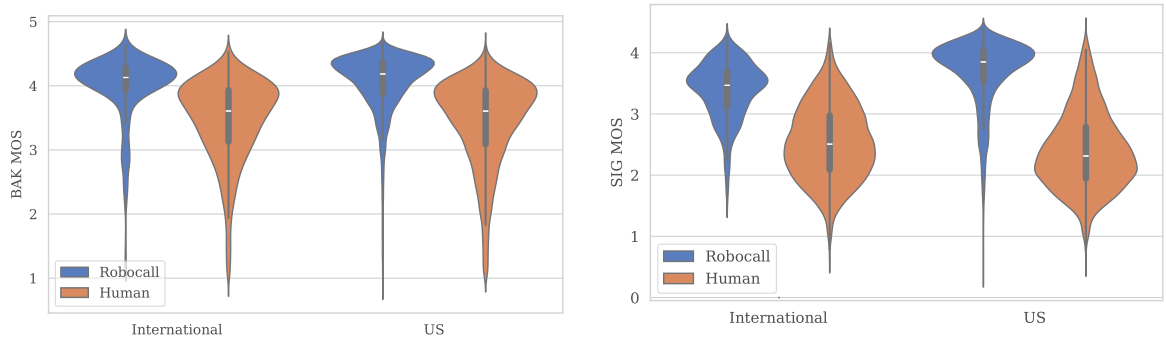
Fig. 17 shows the distribution of DNSMOS (Reddy et al., 2022) metrics SIG and BAK evaluated on international and US calls, split between human calls and robocalls. Expectedly, human calls contain more background noise, which results in a lower BAK score on average. However, the distribution of SIG scores varies even more between robocalls and human calls. We therefore fit a logistic regression to the data, to assess how well the BAK and SIG metric can serve as the sole discriminator between robocalls and human calls. The results are shown in Table 7. The best fit was obtained for US robocalls using SIG as the predictor. The coefficient was $\beta = 2.676$ with a standard deviation of $\sigma = 0.093$, resulting in a z test statistic of $z = 80.90$ and the chi-square-like test statistic $\chi^2(1) = 13{,}554$, both of which are statistically significant ($p \ll .001$). The model's predictive fit is strong with a McFadden's pseudo $R^2 = .485$. We further verified the predictive capability with a cross-validated AUC of $0.923 \pm .003$. Similar conclusions can be drawn for the remainder of US calls and international calls and metrics SIG, BAK, but the predictive capabilities are poorer compared to predicting robocalls in the US using SIG. All p-values were adjusted for multiple comparisons using the Bonferroni correction.

The process was repeated using the group-bootstrap approach and the results are provided in Table 8. The results are similar to those obtained using the non-bootstrapped approach; strong pseudo R^2 values (.301 and .485), odds ratios (15.79 and 14.52), and AUCs (0.923 and 0.861) indicate that the SIG audio quality metric is itself a strong predictor of robocalls, both in US and international settings.

6.5. Spoofed content analysis

We used Spectra-AASIST3 (Borodin et al., 2024; Lab260, 2026) a state-of-the-art speech anti-spoofing model to determine the amount of spoofed speech in robocalls. Our findings are summarized in Table 9, where we show the percentage of spoofed calls vs. bonafide calls per US and international regions, evaluated at different model confidence thresholds. Since Spectra-AASIST3 outputs the probability that a given speech sample is spoofed, we evaluated how modifying the confidence threshold from 0.1 to 0.9 influences the outcome.

From Table 9 we observed that spoofed speech is more common in the US compared to other regions. We verified this with a Pearson's chi-squared test of homogeneity to assess whether the percentage of spoofed speech is different across US and international robocalls. The results are provided in Table 10. We obtained a statistically significant result at all three model confidence thresholds (adjusted for multiple comparisons using Bonferroni correction), but note that the effect size is negligible to small ($\phi \approx 0.065$). The odds ratio, however, is in favor



(a) Distribution of *BAK* (Reddy et al., 2022). (b) Distribution of *SIG* (Reddy et al., 2022).

Fig. 17. Distribution of *BAK* and *SIG* non-intrusive audio quality metrics from DNSMOS (Reddy et al., 2022). Regardless of region, robocalls are slightly less corrupted with background noise compared to human calls, with the signal quality of robocalls being higher as well.

Table 7
Results of logistic regression — evaluating how well non-intrusive audio quality metrics SIG and BAK (Reddy et al., 2022) serve as predictors of whether a call is a human originated call or a robocall.

Region	Metric	β	σ	z	N	χ^2	p-value	R^2	OR [95% CI]	AUC
US	BAK	2.094	0.036	57.91	20,294	5 333	≪.001	.191	8.11 (7.56, 8.72)	0.815 ± .008
International	BAK	2.146	0.103	20.83	4,057	685	≪.001	.149	8.55 (6.99, 10.46)	0.794 ± .017
US	SIG	2.759	0.034	80.90	20,294	13,554	≪.001	.485	15.79 (14.76, 16.88)	0.923 ± .003
International	SIG	2.676	0.093	28.73	4,057	1 417	≪.001	.307	14.52 (12.10, 17.43)	0.861 ± .013

Table 8
Group-bootstrapped results of logistic regression — evaluating how well non-intrusive audio quality metrics SIG and BAK (Reddy et al., 2022) serve as predictors of whether a call is a human originated call or a robocall.

Region	Metric	χ^2 [95% CI]	R^2 [95% CI]	OR [95% CI]	AUC [95% CI]
US	BAK	5,452 (5,449, 5,455)	.194 (.194, .194)	8.42 (8.41, 8.43)	0.820 (0.820, 0.820)
International	BAK	686 (685, 687)	.149 (.149, .149)	8.72 (8.69, 8.78)	0.794 (0.794, 0.794)
US	SIG	13,665 (13,662, 13,667)	.487 (.487, .487)	15.03 (15.02, 15.04)	0.923 (0.923, 0.923)
International	SIG	1,479 (1,478, 1,481)	.321 (.320, .321)	15.50 (15.92, 15.98)	0.867 (0.866, 0.867)

of US robocalls ($OR \approx 1.5$), indicating that US robocalls are ≈1.5× more likely to be spoofed.

The process was repeated using the group-bootstrap approach, which resulted in a statistically significant result at all confidence thresholds ($p \ll .001$), however the odds ratios are of a smaller magnitude, thereby indicating a lower degree of separation between US and international robocalls: threshold 0.1: 1.38 OR (95% CI: (1.27, 1.50)), threshold 0.5: 1.36 OR (95% CI: (1.25, 1.48)), and threshold 0.9: 1.36 OR (95% CI: (1.25, 1.47)).

6.5.1. Callback number extraction

A callback number is a phone number that robocallers share with the callee after the call has been answered (Prasad et al., 2023). This is a straightforward strategy for robocallers. First, they make a call from a spoofed phone number and impersonate a legitimate organization. Second, during the robocall, they share another phone number with the callee, the *callback number*, which is not spoofed, since the robocaller's goal is to gain the target's trust and receive phone calls on the callback number. Usually, receiving a call on the callback number implies that the robocaller has already gained some trust from the callee, as the callee decided to call them. The difficulty with uncovering callback numbers is that they only exist in spoken form, i.e. callback numbers do not appear in metadata, they are only spoken during the call.

We performed callback number extraction by processing call transcripts. The percentage of calls that contain a callback number varies substantially depending on whether the call is US or international: 33.6% of US calls were found to contain a callback number, compared to 1.8% of international calls. For international calls, we found a total of 51 unique callback numbers, compared to 1,513 unique callback numbers in US calls. Four out of the top five callback numbers in the US were confirmed to be scam robocallers by Robokiller (2026), while the fifth was confirmed to belong to a spam/promotional service not listed by RoboKiller. This is aligned with our findings from Section 6.2.3, where we implied that US robocallers are more likely to manipulate the target into calling a specific number, while international robocallers try to deceive the target by making them interact with the robocall.

A Pearson's chi-squared test of homogeneity was performed to assess whether the occurrences of callback numbers within transcripts is

Table 9
Percentage of spoofed speech in US and international robocalls evaluated at different model confidence thresholds using Spectra-AASIST3 (Borodin et al., 2024; Lab260, 2026).

Confidence threshold	US		International	
	Spoofed	Bonafide	Spoofed	Bonafide
0.1	84.42%	15.58%	78.16%	21.84%
0.5	81.02%	18.98%	74.09%	25.91%
0.9	79.30%	20.70%	71.90%	28.10%

Table 10
Results of statistical tests of significance between amount of spoofed speech in US and international robocalls evaluated at different model confidence thresholds.

Confidence threshold	χ^2	p-value	ϕ	Odds ratio
0.1	95.83	≪.001	0.063	1.51 (1.39, 1.65)
0.5	101.31	≪.001	0.064	1.49 (1.38, 1.62)
0.9	108.33	≪.001	0.067	1.50 (1.39, 1.62)

different across US and international calls. The proportion of transcripts that contain callback numbers differs significantly between the two; $\chi^2(1) = 1691.75$, $p \ll .001$, with the difference showing a small to medium effect size ($\phi = 0.263$). With an odds ratio of 28.07 (95% CI: $(22.19, 36.01)$), US transcripts are 28.07 more likely to contain a callback number.

Repeating the process using the group-bootstrap approach confirmed the results: $\chi^2(1) = 1527.63$ (95% CI: (1526, 1530)), with an odds ratio of 29.12 (95% CI: (22.60, 38.09)), thereby confirming that there is a significant difference with a large effect in the ratio of callback numbers between US and international robocalls.

7. Discussion

In this section, we provide a summary of the key findings of our research supported by evidence from the data and address the research questions. We also consider the limitations our study has and provide perspectives and directions for future work.

7.1. Findings

In our study we analyzed 8.7 million CDRs and 220,301 call recordings, collected over a period of 9 months from a joint US and international honeypot.

7.1.1. Call volumes

Our findings indicate that US citizens are substantially more affected by robocalls than the rest of the world. The US honeypot accounted for 11,751 phone numbers compared to 101,913 phone numbers internationally (Table 1). Despite that, US phone numbers received 8,314,813 calls (95.1%) compared to 432,538 (4.9%) calls received by the international honeypot. The average number of calls received per phone number in the US was 692, compared to the international average of 4.25. When analyzed individually per country, the US is still substantially more at risk, as the second country by mean number of received calls is the UK with a mean of 126.9 calls per available phone number. This effect remained statistically significant and practically meaningful when evaluated using the group-bootstrap approach as well. The median calls per available number-day of US to non-US ratio is 134.41. We conclude that US numbers are substantially more at risk compared to non-US numbers.

7.1.2. Calling patterns

Despite the differences in frequency, we found no substantial differences between temporal calling patterns of calls made towards US and international numbers: 61.8% of US-targeted and 69.9% of international-targeted calls are made during weekday business hours (Fig. 7). In both regions, the distribution of calls is heavily skewed: a few callers account for the majority of the calls, with the top 1% of callers accounting for 24.2% of all calls. US calls were found to originate from 501,960 unique caller IDs with a Gini coefficient of 0.66, while international-targeting calls originated from 71,982 unique caller IDs with a lower Gini coefficient of 0.53, suggesting either more distributed operations or less sophisticated automation. With only 119 caller IDs (0.02%) appearing in both US and international CDRs, we can imply that robocall campaigns targeting different regions operate as almost entirely separate operations with distinct infrastructure.

7.1.3. Regional calling patterns

Regarding geographic coverage, the honeypot received calls from 178 unique caller countries to 65 unique callee countries. US numbers account for the majority of callers (85.7%), followed by Nigeria (3.9%), the UK (2.4%), Saudi Arabia (0.8%), and Canada (0.7%). Examining caller–callee country pairs reveals a strong evidence of neighbor spoofing: 89.9% of all calls display a caller country matching the callee country. We found no statistically significant evidence of the matching-country proportions difference between US and international calls. The dominant pair is US to US (85.5% of all calls), UK to UK (2.4% of all calls), and Poland to Poland (0.6%). Cross-border caller IDs appear primarily for US-targeted calls. Nigeria accounts for 3.9% of displayed caller origins, followed by Saudi Arabia (0.7%), Kenya (0.6%), and Afghanistan (0.5%).

The domestic origin was the most frequent caller for 54 out of 65 destination countries, while for 47 destination countries a single dominant external caller can be identified (Fig. 6). The US is the common dominant external source for 16 countries, followed by the UK for 5 countries. Nigeria is a leading external source for calls to the US (340,808 out of 787,594) and it is the dominant external source for a majority of US states (35).

7.1.4. Temporal co-targeting patterns

Our joint analysis of call detail records and transcripts revealed temporal co-targeting patterns consistent with campaign-like organization. First, without considering any transcript content, we constructed an undirected co-targeting graph separately for US and non-US callees, in which each node is a displayed caller ID. We connected two IDs when they called the same honeypot number within a 24-hour window. Most groups obtained using weighted Leiden community detection (Traag et al., 2019) corresponded to complete connected components (99.7% in the US graph and 99.1% in the non-US graph), while Leiden divided a small number of larger components into smaller groups. In other words, displayed caller IDs separate into groups based on timing and shared targets alone, before any content is considered (Fig. 9).

Introducing the transcripts into the analysis provided further evidence of content coherence within these groups. We embedded each transcript into a vector representation using BGE-M3 (Chen et al., 2024) and compared the cosine similarity within and between groups. Within-group similarity was higher by a mean of 0.220 for US groups (95% group-bootstrap CI: $[0.196, 0.245]$) and 0.167 for non-US groups (95% CI: $[0.092, 0.247]$). Within-group transcript similarity is higher on average among eligible groups in both regions (Fig. 10). The results remained consistent across the tested temporal windows, target-exclusion settings, and caller-frequency thresholds, with higher within-group similarity in all tested settings. We therefore interpret the observed content coherence as consistent with campaign-like organization, without attributing the groups to specific campaigns or actors.

7.1.5. Audio content patterns

We analyzed the audio content of 220,301 calls collected in the US and 42,798 calls from the rest of the world. With voice activity detection (VAD) (Silero Team, 2024), we confirmed results from earlier work – a large portion of robocalls are silent calls, i.e. calls without any spoken content (Prasad et al., 2020, 2023, 2025). We also found US calls to contain a substantially smaller portion of voiced calls (24.78% voiced US calls, 41.02% voiced international calls). Although statistically significant, the difference between the vocal content of voiced calls in the US and international calls was negligible from a practical standpoint ($r_{rb} = 0.69$).

In Figs. 13 and 14 we illustrated the risks of hallucinations occurring in automatic speech recognition and confirmed some of the most commonly occurring hallucinations made by Whisper (Radford et al., 2023; Koenecke et al., 2024; Barański et al., 2025). We therefore limited our analysis to calls that contain a sufficient amount of spoken content, similar to previous work (Prasad et al., 2020). Our analysis found English to be the most frequent language in both US and international calls (Table 2) and we analyzed the similarities and differences of the most frequently occurring words in US and international calls for English, Spanish, and Polish (Table 3). Fig. 15 illustrated the distribution of languages per country, with English and Spanish occurring in a number of different countries.

We used DBSCAN (Ester et al., 1996) to identify robocall campaigns based on their transcript and provided the results in Tables 4 and 5. We further applied a few-shot robocall classification and categorization approach by using an LLM (ChatGPT-5.4) to classify robocalls based on their transcript. The LLM then categorized robocalls into 8 distinct categories: *Business and Marketing*, *Financial Services*, *Government Impersonation*, *Health and Insurance*, *Legitimate Automated Notification*, *Political and Non-Profit*, *Technical Support Scam*, and *Unknown/Other*. These results were compared and anchored against labels assigned by a human annotator and another LLM, where we found a relatively high-degree of inter-annotator agreement, but concluded that it is category-dependent. The exact distribution of robocalls per category per region is provided by Fig. 16. A chi-square test of homogeneity indicated that there is a significant difference in robocall categories between US and international robocalls. A post-hoc analysis revealed that *Financial Services* and *Political and Non-Profit* robocalls are substantially more common internationally, compared to *Business and Marketing*, *Health and Insurance*, and *Technical Support Scams* that are more frequent in the US, with *Legitimate Automated Notifications* being a significantly more common means of communication in the US compared to the rest of the world. The practical relevance of these differences should, however, be analyzed per-category, as for example the differences in some categories are statistically significant (e.g. Technical Support Scam), but not very meaningful, while for some categories (e.g. Financial Services) the differences are substantial.

Audio quality was analyzed using DNSMOS P.835 (Reddy et al., 2022), a non-intrusive audio quality predictor that predicts the quality of an audio sample measured across three variables: SIG, BAK, and OVRL. We found that the metric SIG can serve as a good predictor of whether a call is a robocall, especially on US calls where we obtained a cross-validated AUC of $0.923 \pm .003$ (Table 7 and Fig. 17). These results were reaffirmed to be persistent across different displayed caller IDs by applying the group-bootstrap approach.

Spectra-AASIST3 (Borodin et al., 2024; Lab260, 2026) was used to determine the amount of spoofed speech in robocalls. We obtained a statistically significant result at different model confidence thresholds (Tables 9 and 10) with US robocalls being about 1.5 times as likely to contain spoofed speech compared to international robocalls.

Callback number extraction indicated that 33.6% of US robocalls contained a callback number compared to 1.8% of international calls. With a statistically significant difference in the distribution of callback numbers across robocalls, US robocalls were 28.07 times as likely to contain a callback number compared to international robocalls. Using Robokiller (2026), we confirmed that 4/5 top callback numbers were confirmed to be scam robocallers, with the fifth robocaller found to be a spam/promotional service as well.

7.2. Perspectives and future directions

Our findings showed that US robocalls differ from robocalls in the rest of the world: they are substantially more frequent, more likely to be silent, more likely to contain spoofed speech, and more likely to contain a callback number. We found temporal co-targeting groups with similar call content in both US and non-US traffic.

Here, we want to address speculative claims that would need to be further verified and that could serve as basis for future research. Given that US robocalls are more likely to be silent and given the substantial drop (up to tenfold) in call volume once the honeypot switched from passive to interfering mode (Fig. 3), we have reason to believe that robocallers are using silent calls to simply identify which phone numbers are active and capable of answering a phone call. Another plausible explanation could be that callers are using voice activity detection features that trigger playback of recorded messages once, and only if the caller is confident that the call has been answered by an actual person. This method is known as *answering machine detection* (AMD) (Gómez et al., 2014; Altwlkany et al., 2024) and it has been used by robocallers before (Tu et al., 2016). It would be interesting to compare the difference in silent calls between calls that were answered with a warning message and calls answered with a prerecorded human message. This would, however, require a stricter legal and ethical assessment since callers would not be warned about the recording taking place.

An interesting observation was that the words *call* and *llamada* were the most frequent words in English- and Spanish-language US calls, respectively, but were not the most frequent in international calls in those languages. On the contrary, the words *press* and *oprima/presione* were top-occurring words in international calls in English and Spanish. This finding, together with the following findings for US calls: (1) larger amount of spoofed calls, (2) substantially higher amount of callback numbers, and (3) larger percentage of silent calls gives us reason to believe that US robocallers are more sophisticated and rely more on advanced technology and have a more complex robocalling procedure. We believe that they are more inclined to convince the target victim to call (*llamada*) a callback number. Additionally, we found that Legimate Automated Notifications are substantially more common in the US. This might imply a cultural difference that partially explains the disproportionate severity of malicious robocalls in the US compared to the rest of the world, namely, US citizens are used to receiving legitimate notifications through the telephone network, while this is not the case in other countries.

Furthermore, given our data, the US dominates the number of unsolicited calls, but the UK substantially deviates from the international average as well. It would be interesting to consider an independent study of robocalls in the UK and other English speaking countries at a larger scale.

7.3. Limitations

Working with robocalls is a sensitive topic, from both a legal and ethical perspective. We tried our best to preserve caller privacy and ensure no legitimate calls were processed during our research. This, however, resulted in our study introducing several limitations, such as playing a warning message whenever calls were answered or omitting caller IDs that made only a single call to the honeypot. The number of transcripts (839) and audio recordings (677) released in the dataset is substantially lower than the number of CDRs (8.7 million), as they require manual confirmation by human annotators, often in multiple languages. The co-targeting analysis is limited by sparse transcript coverage, with usable transcripts available for 0.68% of US and 0.89% of non-US graph nodes; therefore, the findings apply only to transcript-covered groups and do not identify specific campaigns or actors.

7.3.1. Honeypot limitations

The honeypot itself featured several unresolved limitations that were partially addressed in Section 4. First of all, a technical limitation of the honeypot had caused the quantity of available phone numbers to vary slightly per day. Second, the number counts per country (Table 1) reflect the honeypot provider's number availability in each market, not a stratified or population-proportional sample, implying a bias.

To quantify how sensitive our US vs. international comparison is to this non-representativeness of the honeypot, we ask how large of a differential representativeness bias between the two groups (US and international) would be needed to produce the observed gap by artifact alone. More precisely: if numbers from our honeypot of a given country or collection of countries c receive calls at rate ρ_c times the true/typical rate that representative in-use numbers would receive in c, our calls per available number-day ratio is unbiased for the true exposure ratio only if $\rho_{\mathrm{US}} = \rho_{\mathrm{int}}$. The observed ratio was found to be 134.41 (95% CI: (106.22, 164.47)) which means that for this effect to be a product of chance (an artifact), the US honeypot numbers would have to be $\approx$134 less representative of true US call exposure than international honeypot

numbers are of their true call exposure. Even at the conservative CI interval this yields a ≈106 asymmetry. We have no evidence of such an asymmetry as the honeypot numbers in both groups were acquired by the same honeypot provider using the same process. The variety of countries in the international part of the honeypot offers more evidence; it spans 64 countries acquired under the same non-representative process, with heterogeneous national numbering plans and carrier practices that would plausibly produce their own representativeness anomalies if the effect size required here were realistic. The largest non-US deviation from the international mean (4.25) was observed in the UK (126.9), a ≈30-fold deviation, which is well below that of the ≈134-fold asymmetry that would be needed between the US and the rest of the world for the gap to be an artifact. We therefore consider it unlikely that the number availability representativeness bias explains the magnitude of the US vs. international gap, although we cannot rule it out directly and this remains an unresolved limitation of our work. The same argument is considerably weaker for the ranking of individual non-US countries against each other, so country-level rankings should accordingly be considered as more sensitive to this limitation than the aggregated US vs. international claim.

The final unresolved limitation of our honeypot is the technical limitation which featured two periods of inactivity during which no calls were registered. We wish to at least to some degree quantify the extent to which this impacts the result. We consulted a proprietary but publicly available robocall index that contains historical data from 2016 to 2026 (YouMail, 2026). When aggregated over this 10-year period, 8.06% of robocalls were made in the month of February (lowest), with the most robocalls being made in the month of October (8.87%). The months which were affected by the honeypot inactivity are: June (8.23%), July (8.13%), August (8.55%), and November (8.66%). Based on this robocall index, robocalls were historically distributed relatively evenly across months, given that an exact uniform spread would amount to 8.33% of robocalls per each month. We note that this analysis is based on just one publicly available robocall index with a historically large coverage of robocalls. The idea here is not to claim that the limitation can be ignored, rather to at least quantify it to some degree.

7.3.2. Multiple hypotheses testing

Statistical tests were grouped into distinct families based on the core outcome measure that was evaluated (e.g. call duration, presence of callback numbers, spoofing probabilities, etc.). To control for Type I error inflation, we adjusted the p-values using either the Bonferroni or Holm–Bonferroni correction method independently within each distinct family of hypotheses, the choice of which is immaterial given the p-value magnitudes.

Despite the application of family-wise corrections, the overall number of statistical tests across the study increases the cumulative probability of at least one false positive. Consequently, secondary findings should be interpreted with appropriate caution and validated in future work.

8. Conclusion

In this paper, we presented a detailed overview of the state of international robocalls. Our honeypot registered 9.6 million calls, of which we make publicly available a dataset of 8.7 million call detail records, 677 robocall recordings, and 839 clustered and categorized robocall transcripts. Our analysis showed that although robocalls are an issue in other countries, the threat severity is highest in the US. We analyzed temporal and geographical calling patterns and examined co-targeting behavior within short time windows. By analyzing call content, we showed the most frequently occurring languages in robocalls and robocall scripts and their (dis)similarities based on region. We used an unsupervised clustering approach to identify frequent robocall campaigns, as well as an LLM-based approach that categorized robocalls into common categories via few-shot prompting. We estimated the percentage of spoofed content within robocalls and found that robocalls can be distinguished from human originated calls by considering non-intrusive audio quality metrics. For future work, we plan to extend our dataset and overcome some of the technical limitations of our study, as well as to work on directly combating robocalls.

CRediT authorship contribution statement

Kemal Altwlkany: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Supervision, Conceptualization. **Andro Merćep:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Tomislav Đuričić:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Ante Kapetanović:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Emanuel Lacic:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Data curation, Conceptualization.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude Code (Opus 4.6) for code implementation and data analysis, ChatGPT 5.2 for LaTeX editing and grammar checking, and Google Scholar Labs for related-work identification. After using these tools and services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Reports of telephony spam and fraud

In the Introduction section, we mentioned that robocalls are on the rise in the US: according to the report of the U.S. Public Interest Research Group Education Fund (2025), robocalls are reaching a 6-year high monthly average of 2.56 billion, as was shown in Fig. 1. This roughly translates to US citizens receiving 7.5 monthly robocalls per person, on average. These robocalls are not just harmlessly bothering citizens, as US citizens have reported a total of $1.1 billion lost to fraud over phone calls in 2025, according to the report of the FTC (Federal Trade Commission Public Tableau, 2025a). The same report lists that

phone calls are the third most reported method of contact for fraud at 17%, right behind text and email. However, the median amount of money lost to fraud committed over the phone peaks at $1,835. For comparison, second on this list is text, for which the median money loss is $1,000, which is almost double the difference. The 2025 statistics are not an anomaly, for example, reported median money losses in 2022 were also highest for telephony fraud, with the median reported loss over phone call scams amounting to $1400, according to the FTC (Federal Trade Commission, 2023).

The Federal Bureau of Investigation (FBI) founded its Internet Crime Complaint Center (IC3) in 2000. During its infancy, IC3 received roughly 2000 complaints every month (Federal Bureau of Investigation IC3 Division, 2025, 2026). From 2019 to 2024, the complaints were averaging more than 2000 per day. Losses reported to the FBI are on the rise: a total of $16.6 billion in losses was reported in 2024 across several types of fraud with phishing and spoofing topping the list by number of complaints at more than 190 thousand, more than twice as much as extortion, the second on the list at over 86 thousand complaints (Federal Bureau of Investigation IC3 Division, 2025). In 2025, a record $20.9 billion in total losses was reported, with phishing and spoofing topping the list again with a record-high 191 thousand complaints.

The FTC's Do Not Call (DNC) registry, as of 2025, contains 258 million active DNC registrations. More than 2.6 million complaints were made during 2025, of which 61.2% were robocalls (Federal Trade Commission Public Tableau, 2025b). These numbers are on the rise: in 2024 the number of complaints was 2.09 million with 52.7% robocalls; in 2023 a total of 2.12 million complaints, with robocalls contributing 54.6%.

Interested readers are advised to consider the FCC's 2024 report, in which the FCC addresses the uses of artificial intelligence to protect consumers from unwanted robocalls, robotexts, and other harms, as well as enabling people with disabilities to make calls (Federal Communications Commission Consumer Advisory Committee, 2024).

A.1. A US-only problem?

A lot of attention has been devoted to the US, which we attribute to the following reasons:

1. Previous research revolved around the US, making it appealing to focus on the US so that comparisons can be drawn more easily against existing baselines. Previous research also influenced the availability of data.
2. The US has a very broad number of transparent government and non-government organizations which contribute to the amount of publicly available data, making it easier to work with robocalls.
3. As the data from our paper confirms as well, US citizens are by far the largest target of robocallers. Even though other countries are affected by robocalls, the degree of robocall quantity and severity is unparalleled compared to the US. However, this does not mean that other countries are exempt from robocalls, or that the problem itself is insignificant.

A.2. Africa

The Global Anti-Scam Alliance (GASA) 2025 report found that 68% of adults in surveyed African countries encountered at least one scam, with **Kenya** (83%), **South Africa** (77%), and **Nigeria** (73%) being the most affected countries (Centre for Multilateral Affairs, 2025). Voice calls contributed to 51% of the scams.

Kenya's Communications Authority reported 2.54 billion cyber threat incidents in Q1 2025, a 201.7% increase from the previous quarter. The Computer Misuse and Cybercrimes (Amendment) Act of 2025 expanded the definition of phishing offences to explicitly cover fraudulent phone calls, broadening the scope beyond text-based digital communications (Vellum Kenya, 2025).

INTERPOL's Operation Red Card (2024) (INTERPOL, 2025) resulted in over 300 arrests throughout Africa. In **South Africa**, 40 individuals were arrested and more than 1,000 SIM cards were seized. According to a 2025 report from the Communication Risk Information Centre (COMRiC) an estimated $300 million was linked to telecom fraud in South Africa, costing the country close to 1% of its GDP (Connecting Africa, 2025).

In January 2025, 105 suspects were arrested, including 4 Chinese and 101 Nigerian citizens in Abuja, **Nigeria**. The fraudsters were targeting victims in Europe, primarily the United Kingdom (Economic and Financial Crimes Commission, 2025). A month earlier, in December 2024, the Economic and Financial Crimes Commission (EFCC) had performed a raid, arresting 792 suspects for their alleged involvement in cryptocurrency investment fraud and romance scams, of which many of them international citizens.

A.3. Asia-pacific and middle east

According to the **Japan** National Police Agency (NPA), the total amount of money lost to unsolicited phone calls and social media reached its highest level ever in the first half of 2025 (Tokyo Reporter, 2025). Fraudsters have collected close to ¥59.7 billion ($380 million) between January and June 2025, which is 2.6 times higher than in the same period the year before. International calls accounted for approximately 70% of fraud calls in 2024, rising to 80% in the first half of 2025 (The Japan Times, 2025). Over the whole of 2025, ¥141.42 billion ($900 million) in losses was attributed to telephony fraud, with victims typically being contacted via international phone calls and then directed to video chats on the messaging app Line (The Japan Times, 2026).

Voice phishing losses in **South Korea** hit a five-year high of ₩433.8 billion ($317 million) in 2024, up 14.1% from the previous year, while per-case damage doubled to approximately ₩19.9 million (Seoul Economic Daily, 2026). SK Telecom, South Korea's largest carrier, reported blocking 250 million voice spam and scam calls in 2025 (Commsrisk, 2025). INTERPOL's Operation HAECHI V (INTERPOL, 2024) in 2024 targeted seven types of cyber-enabled fraud, including voice phishing. Korean and Chinese authorities dismantled a voice phishing syndicate responsible for ₩1,511 billion ($1.1 billion) in losses affecting over 1,900 victims, resulting in 27 arrests.

China. The Ministry of Public Security reported intercepting 2.75 billion fraudulent phone calls and 2.28 billion suspicious SMS messages, and stopping 13.89 million people from being defrauded in 2023 (China Daily, 2024). From January to November 2024, more than 67,000 individuals were indicted on telecom fraud charges, a 58.5% increase year-on-year (Supreme People's Procuratorate of China, 2025). China adopted its own national caller identification standard (GB/T 43779-2024) (Commsrisk, 2024), published in April 2024 with an implementation date of 1 November 2024, as an alternative to the previously mentioned US-originated STIR/SHAKEN (Peterson et al., 2014; Peterson and Turner, 2018; Rescorla and Peterson, 2021; Wendt and Barnes, 2019).

The oldest television broadcaster in Southeast Asia, ABS-CBN, released a report in which they claim that the number of SMS scams in the **Philippines** has decreased in 2025, but that the number of scam calls rose by 78% in Q3 of 2025 (ABS-CBN News, 2025).

In **Australia**, the Australian Competition and Consumer Commission (ACCC) reported that in 2023, Australians lost over $3.1 billion to scams, with phone scams being a significant contributor (Pietri et al., 2025).

Singapore and **Malaysia** signed a Memorandum of Understanding in 2024 to enhance cooperation against telecommunication scams (Pietri et al., 2025; Indiplomacy, 2026).



Fig. 18. Info flyer published by the Croatian Police as part of Europol's campaign relating to phone theft and fraud. If the callers present themselves as an organization the police instructs citizens to not call back numbers that the caller might have specified during the call. Citizens are instead advised to look up official phone numbers of those organizations. This info flyer was automatically translated using Google, the original can be found on the Croatian Police's website (Republic of Croatia, Ministry of the Interior, n.d.).

The FBI, specifically the IC3 department, issued a public service announcement in May 2025 warning of fraud schemes targeting Middle Eastern students (from the **UAE**, **Saudi Arabia**, **Qatar**, and **Jordan**) studying in the United States, where scammers impersonate government or immigration officials and threaten deportation; government impersonation fraud cost Americans over $405 million in 2024 (Federal Bureau of Investigation, 2025).

UAE's largest telecom, e&, has reported rolling out AI-based software to combat robocalls (Andrew Wooden, 2025). The **Saudi Arabia** Data and Artificial Intelligence Authority has issued a warning for consumers to reject telephone calls from people seeking their personal or bank details, warning users about vishing attempts (Arab News, 2022).

A.4. South America

Effective 13 August 2025, all commercial calls in **Chile** must carry mandatory prefixes: 600 for solicited calls (from entities with which the consumer has an existing relationship) and 809 for unsolicited telemarketing or advertising calls (Subsecretaría de Telecomunicaciones de Chile, 2025; IDT Express, 2025). The regulation, enforced by the Subsecretaría de Telecomunicaciones (Subtel), enables consumers to instantly distinguish legitimate calls from potential spam before answering.

According to Hiya's 2025 report (Hiya, 2025), **Brazilians** receive on average 28 spam calls per month, close to 1 spam call per day, with 17% of those calls estimated to be fraud calls.

According to Truecaller's annual report for 2025 (Truecaller, 2026), on average 6.8 calls out of 10 calls received by Brazilians outside of their phone contact list are spam, with the ratio being slightly higher in Chile at 7/10.

Argentina's National Entity of Communications (ENACOM) enforces regulations that require telecom operators to implement systems for detecting and blocking spam calls and messages (Pietri et al., 2025).

A.5. Europe

In **Croatia**, numerous news articles have reported incidents of fraud performed via phone robocalls. The Croatian Police (Republic of Croatia, Ministry of the Interior, n.d.) published flyers instructing citizens how to behave if they receive unexpected phone calls, an example of which is shown in Fig. 18.

The Netherlands. NOS, part of the Dutch public broadcasting system, released an article in which they state that the Dutch Fraud Help Desk received nearly 10,000 reports of scam calls that try to defraud people and extort money (NOS, 2025). Scammers are relying on AI-enabled technology to facilitate such calls.

The news portal Telecoms reported that O2, the largest telecom in the **United Kingdom**, detected 150 million suspected scam and spam calls and blocked millions more that were later confirmed as fraudulent (Nick Wood, 2025). More than 50 million suspected scam calls are flagged every month with the help of AI-based detection software, the details of which were not disclosed.

Spain's largest telecommunication company Telefonica stated that they are blocking almost half a million spam or fraudulent calls daily in 2026 (Telefonica, 2026).

Since 19 November 2025, **Italian** telephone operators have been rejecting on average 7.5 million calls every day that are disguised as Italian numbers, but actually originate from abroad (Il sole 24 Ore, 2025).

Data availability

The dataset supporting this study is publicly available on Zenodo: *International Robocalls Dataset*, version 2 (Altwlkany et al., 2026). The dataset is released under the Creative Commons Attribution-Non Commercial 4.0 International license.

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